

Occupant-centric HVAC and window control: A reinforcement learning model for enhancing indoor thermal comfort and energy efficiency

Xin Liu, Zhonghua Gou^{*}

School of Urban Design, Wuhan University, Wuhan, China

ARTICLE INFO

Keywords:

HVAC and window control
Occupant behavior
Reinforcement learning
Thermal comfort
Energy efficiency

ABSTRACT

Occupant behavior plays a crucial role in enhancing indoor thermal comfort and achieving energy efficiency by influencing the operational modes of Heating, Ventilation, and Air Conditioning (HVAC) systems as well as windows. However, accurately quantifying the impact of occupant behavior on the indoor environment presents significant challenges in practical applications. This study introduces an innovative approach by leveraging the ASHRAE Global Building Occupant Behavior Database and harnessing the power of XGBoost in conjunction with Deep Q Networks (DQN) to construct a reinforcement learning model. This model enables precise prediction of the impact of occupant behavior on the indoor environment at the next time step under varying indoor-outdoor conditions, simultaneously targeting the dual objectives of indoor thermal comfort and energy conservation. By applying the XGB-DQN model in sample rooms of four international cities with distinct features, the results demonstrate a significant increase in indoor thermal comfort duration by 24 %, accompanied by a 24.7 % decrease in air conditioning usage compared to baseline models and actual occupant data. This research represents a pioneering effort in applying reinforcement learning techniques to accurately predict occupant behavior's impact on indoor environments, offering valuable insights for intelligent building design and energy management.

1. Introduction

The primary means of regulating indoor thermal conditions, namely Heating, Ventilation, and Air Conditioning (HVAC) systems, stand out as the principal source of building energy consumption. According to the 2022 Global Building Performance Report [1], energy consumption during the operational phase of buildings represents over 30 % of global energy consumption. Notably, HVAC or central air conditioning systems contribute significantly to the highest energy consumption in the operation of public buildings, accounting for 60 %. The central focus in the field of building energy efficiency is on addressing the challenge of reducing the energy consumption of air conditioning systems while maintaining indoor thermal comfort [2]. Optimizing the operation mode of HVAC systems for environmental control can lead to substantial energy savings by reducing unnecessary HVAC operation while ensuring thermal comfort for occupants through various control measures [3].

Numerous studies have proposed control methods based on building thermal comfort control and energy optimization, predominantly employing Rule Based Control (RBC), Proportional Integral Derivative (PID), Model Predictive Control (MPC), and Reinforcement Learning

(RL) [4]. Over the past decade, reinforcement learning (RL) has seen the emergence of many effective learning algorithms, gaining increasing application in research areas such as building energy systems, including heating, ventilation, and air conditioning [5]. In comparison to other traditional learning approaches, RL does not necessitate the construction of a thermodynamic model of the system, presenting a significant advantage in addressing complex, dynamic, and highly uncertain issues in HVAC systems. Moreover, RL enables the evaluation of both short-term and long-term rewards on decisions and can adapt to different buildings and climates through self-adjustment methods such as simulation or actual operation [6].

The essence of reinforcement learning lies in facilitating ample interaction experiments between the Agent and the Environment, learning the optimal strategy from a vast array of interaction samples. However, in many practical applications, the dynamic changes in the environment are unknown and challenging to model. Direct experimentation on the real environment can result in equipment loss and resource wastage. A common strategy to address this challenge is to integrate reinforcement learning with supervised learning methods such as Recurrent Neural Networks (RNN) and EXtreme Gradient Boosting (XGBoost), effectively mitigating the instability of reinforcement

^{*} Corresponding author.

E-mail address: zh.gou@whu.edu.cn (Z. Gou).

<https://doi.org/10.1016/j.buildenv.2024.111197>

Received 2 December 2023; Received in revised form 7 January 2024; Accepted 11 January 2024

Available online 14 January 2024

0360-1323/© 2024 Elsevier Ltd. All rights reserved.

Nomenclature*Latin symbols*

$\hat{y}_i$	Predicted value of y
$\mathcal{L}(\varphi)$	Objective function
$l(y_i, \hat{y}_i)$	Loss function
$\Omega(f)$	Regularization term
g_i	First-order derivative
h_i	Second-order derivative
S_t	Status space
A_t	Action space
R_t	Total reward
β	Coefficients for determining working hours
ω	Weights assigned to objectives
R_{comfort}	Thermal comfort reward
R_{Energy}	Energy-saving reward
λ	Amplification factor
t_{HVAC}	Unit operating time of HVAC system
P	Rated power of HVAC system
$[T_{\text{lower}}, T_{\text{upper}}]$	Thermal comfort zone
t_{out}	Outdoor temperature
γ	Discount factor
α	Learning rate
$Q(s, a)$	Q network
$Q'(s, a)$	Target Q Network

Acronyms

HVAC	Heating, Ventilation, and Air Conditioning
DQN	Deep Q Network
XGBoost	eXtreme Gradient Boosting
SHAP	Shapley Additive Explanations
DDPG	Deep Deterministic Policy Gradients
MAE	Root Mean Square Error
RMSE	Root Mean Squared Error
RBC	Rule Based Control
MPC	Model Predictive Control
FNN	Fuzzy Neural Network
RNN	Recurrent Neural Networks
RF	Random Forest
KNN	K-Nearest Neighbors
ANN	Artificial Neural Networks
ASHRAE	American Society of Heating, Refrigerating and Air-Conditioning Engineers
PMV	Predicted Mean Vote
PPD	Predicted Percentage Dissatisfied
PMOT	Prevailing Mean Outdoor Temperature
NV	natural ventilation
AC	air conditioning
MM	Mix Mode buildings
SAC	Split Air Conditioned

learning in the early stages of training [7]. However, constructing a high-performance model demands a substantial amount of high-quality data, and its adaptability and generalization ability may be limited in different building environments. Recent research also indicates that using simulation environments for training, such as constructing virtual HVAC system environments in software like Energy Plus, has yielded promising outcomes [8]. On the other hand, some scholars have adopted a comprehensive framework that combines simulation models and supervised models, employing methods like Random Forest (RF), K-Nearest Neighbors (KNN), and Artificial Neural Networks (ANN) to train the simulation environment, presenting a prospective solution [3]. However, differences between virtual and real environments necessitate careful consideration to verify their accuracy.

To address the limitations, this study utilized the ASHRAE (American Society of Heating, Refrigerating and Air-Conditioning Engineers) Global Building Occupant Behavior Database [9]. This collaborative database involves contributions from researchers representing 39 institutions worldwide and offers several advantages. This study utilizes the database to develop a reinforcement learning model that combines XGBoost with Deep Q Networks (DQN) for intelligent regulation of the HVAC-window system. The objective is to achieve simultaneous improvements in thermal comfort and energy efficiency. XGBoost is employed to predict the impact of HVAC-window operation on the indoor environment under varying indoor and outdoor conditions. The DQN agent explores different operation strategies, receives feedback from the environment, and learns optimal control strategies through trial-and-error. By integrating and iterating XGBoost and DQN, the model can accurately compute environmental changes based on initial features and continuously update the environmental state, resulting in optimal HVAC-window control strategies. The study also compares the control strategies generated by the XGB-DQN model with baseline models and occupant-initiated control models, demonstrating the potential of the XGB-DQN model in enhancing indoor thermal comfort and energy efficiency.

2. Literature Review

2.1. Indoor thermal comfort standards

Thermal comfort is usually defined as “that condition of mind that expresses satisfaction with the thermal environment” [10]. In the evaluation of thermal comfort in buildings, two classical models are frequently employed. One is the Predicted Mean Vote (PMV) and Predicted Percentage Dissatisfied (PPD) thermal comfort model proposed by Fanger [11], which comprehensively considers factors such as Air temperature, Humidity, Air speed, Radiant temperature, Clothing insulation, and Metabolic rate to evaluate the thermal comfort of individuals in a given environment [12]. This evaluation model has been widely adopted by international standards, such as ISO 7730 in 1984 and ANSI/ASHRAE Standard 55 in 1992 [13]. However, several on-site studies have revealed significant disparities between PMV values and individuals’ actual thermal sensations in everyday thermal environments, particularly in naturally ventilated buildings with ample opportunities for adaptation [14]. In such scenarios, the temperature range to which occupants can adapt exceeds the initially expected range [15].

The model of adaptive thermal comfort is proposed to address the limitation [16]. The optimal indoor temperature should be primarily linearly related to the outdoor environment. It was officially incorporated into the 2004 version of the ANSI/ASHRAE Standard 55 [17], utilizing the monthly mean outdoor temperature as the reference for outdoor conditions. In the 2017 version of the ANSI/ASHRAE Standard 55, the acceptable range of operative temperatures is divided into two categories: 80 % acceptable and 90 % acceptable [10]. And it explicitly stated that the adaptive model is applicable only to occupant-controlled naturally conditioned spaces. Prevailing Mean Outdoor Temperature (PMOT) is proposed through weighted averaging outdoor temperature [18]. PMOT should range between 10 °C and 33.5 °C; exceeding this range necessitates the installation and operation of mechanical cooling or heating systems [10].

Admittedly, relying solely on natural ventilation has increasingly proven inadequate in providing sufficient thermal comfort for occupants in buildings [19]. Studies indicate that, as the climate warms, there

would be a reduction in annual window opening for 40 % of the population [20]. An increasing number of buildings are integrating natural ventilation (NV) with air conditioning (AC) systems to engender mixed-mode/hybrid-mode ventilation strategies. The 2020 revision of ASHRAE Standard 55 specifies that these Mix Mode (MM) buildings adopt the PMV/PPD thermal comfort model in AC mode and the adaptive model in NV mode [21]. Luo et al. discovered that, compared to steady-state PMV models, adaptive comfort theory better predicts the thermal sensations of residents in hybrid-mode buildings [22]. Yan et al. suggest that residents in AC and NV modes in Split Air Conditioned (SAC) buildings have equal opportunities and abilities to adapt to the current indoor thermal environment, allowing for the use of adaptive theory in evaluations [23]. It is also pointed out that while the neutral temperature increases with outdoor temperature, residents' adaptability is limited when the outdoor temperature exceeds 30 °C. Yun et al. proposed an adaptive comfort control (ACC) strategy for air conditioning systems, reaching similar conclusions. The results indicate that when the outdoor temperature is below 10 °C, the optimal comfort temperature remains unchanged with outdoor temperature variations [24].

Based on the cumulative experiences discussed earlier, this study defines an extended adaptive comfort zone (Fig. 1) for AC-NV systems. The 80 % and 90 % acceptability limits are derived from the ANSI/ASHRAE Standard 55, with outdoor temperature boundaries set within the interval of 10 °C–30 °C. Beyond this boundary range [25], the indoor thermal comfort temperature will no longer exhibit a linear correlation with outdoor temperature.

2.2. Data-driven HVAC control

Numerous studies have endeavored to utilize technologies such as sensors, infrared, and image recognition in real building environments to collect behavioral data from users under different conditions and aimed to infer user needs and consequently design data-driven solutions [26–28]. In the realm of HVAC control research, three predominant methods are Rule-Based Control (RBC), Model Predictive Control (MPC), and Reinforcement Learning (RL) [8].

RBC approaches typically rely on a set of predefined empirical rules, providing a relatively straightforward and intuitive framework. Fiorentini et al. proposed a rule-based control strategy for Mixed-mode buildings, wherein window openings are adjusted based on adaptive

thermal comfort standards, and heating or cooling devices are activated in response to the PMV index [29]. Firlag et al. devised five distinct control algorithms, encompassing heating/cooling, simple rules, perfect citizen, heat flow, and predictive weather, aiming to achieve energy-efficient control in diverse scenarios with varying window configurations [30]. Nevertheless, with the increasing complexity of HVAC systems and building environments, RBC methods encounter challenges in adapting to intricate and dynamic systems.

MPC proves well-suited for multi-variable systems and adept at handling constraint conditions. It engages in optimization at each time step using a system model, forecasting the system's future behavior and selecting the current optimal control inputs. Pandey et al. developed an embedded system based on Raspberry Pi-Arduino, utilizing PMV as a constraint and employing real-time co-simulation with Energy Plus and Simulink to predict the set points of the AC [31]. Kim et al. proposed an MPC solution based on the lumped output disturbance modeling approach, capable of controlling multiple rooftop packaged units in medium-sized commercial buildings to maintain thermal gains while minimizing energy consumption [32]. However, MPC has its limitations, such as being highly dependent on the construction of thermodynamic system models, often deviating from real-world scenarios. Additionally, simulation under complex and dynamic conditions necessitates robust computational resources [33].

RL has garnered more attention than ever before [34]. Appendix A summarizes recent studies in this area. The objective of RL is to enable an intelligent agent to learn how to take actions to maximize cumulative rewards through interactions with the environment [35]. In RL, the agent continually observes the environmental state, executes actions, receives feedback on changes in the environment, computes rewards, and archives this entire process as experiential knowledge [36]. Drawing insights from accumulated experiences, the agent adjusts and refines its behavior, progressively acquiring an optimal strategy [37]. RL's appeal as an alternative to MPC stems from its flexibility, which eliminates the need for stringent physical models, relying instead on the interaction between actions and the environment for learning control strategies [5].

Three main modeling approaches exist for RL: (1) direct interaction with the real environment, (2) simulation and optimization based on software modeling and generated data, and (3) the establishment of state transition models based on historical data from real-world. Traditional RL is an online learning method based on direct

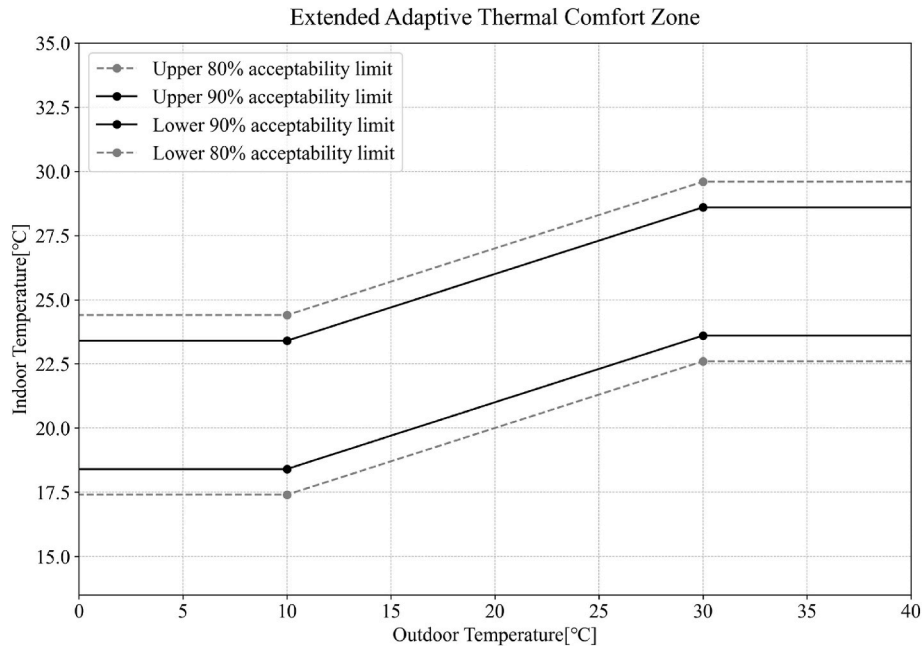


Fig. 1. Extended adaptive thermal comfort zone.

interaction with the real environment, optimizing strategies through experiential learning in a trial-and-error process. However, direct interaction with the real environment in the actual operational context of buildings poses drawbacks such as resource wastage, occupant discomfort, and prolonged time consumption [38]. Many scholars adopt simulation-based methods using software modeling to set control strategies [39]. Deng et al. proposed a novel HVAC control method combining active detection of building environment changes and Deep Q Network (DQN), achieving energy savings of up to 13 % and thermal comfort improvements of 9 % through Energy Plus simulations in commercial building thermal zones [8]. Gao et al. simulated HVAC and thermal states in buildings, predicting thermal comfort using Fuzzy Neural Network (FNN) [40].

Nevertheless, due to the current lack of robust behavioral models in simulation packages, many researchers are attempting to enhance simulation accuracy by leveraging field data collected from real buildings to develop useful behavior models, including HVAC and window behavior models [41]. An et al. collected monitoring data from real rooms over a three-week period and, based on deep reinforcement learning, devised a controller capable of simultaneously managing windows, air purifiers, and air conditioning in authentic environments [42]. This approach resulted in a 16 % improvement in thermal comfort and a 23 % reduction in energy consumption. Lu et al. utilized the ASHRAE RP884 data set, employing models such as KNN/SVM to predict indoor thermal comfort PPD based on indicators like indoor and outdoor temperatures and wind speed and proposed a Q-learning algorithm for temperature set-point control [43]. Han et al. developed a model-free RL method, using LSTM-RNN to predict indoor temperatures and simulate building environments for controlling windows to enhance indoor comfort [7]. Man Wang et al. compared the performance of XGBoost and LSTM in various RL algorithms, presenting a control logic that combines XGBoost with DDPG, which achieved a 10.06 % energy savings while ensuring comfort [33].

In sum, research exploring the application of reinforcement learning to regulate building environments has prioritized optimizing multiple facets, including thermal comfort and energy efficiency. Furthermore, an increasing number of studies are incorporating machine learning predictive models during the development of these models. However, there are still opportunities for improvement. Firstly, the majority of research is localized to specific cities or regions, leaving the transferability and generalizability of reinforcement learning models across cities with diverse climate types unexplored. Additionally, current research tends to focus predominantly on HVAC systems for control, overlooking the collaborative interactions between HVAC systems and other building components, such as windows, in optimizing the building environment.

2.3. Objective

Building on the findings and limitations identified in previous research, this study constructs a reinforcement learning model that integrates XGBoost and Deep Q Learning to intelligently control building HVAC and window systems, with the aim of achieving simultaneous objectives of indoor thermal comfort and energy savings. During the reinforcement learning training process, the study utilizes the ASHRAE Global Building Occupant Behavior Database On-site data set and compares the performance of the reinforcement learning model across multi-cities: mainly Rende, Nanjing, Sao Paulo, and Florianopolis. The study posits forth the following hypotheses.

H1. XGBoost demonstrates strong performance in predicting the impact of occupant behavior on indoor environments.

H2. The reinforcement learning model, combining XGBoost and Deep Q Network, enhances indoor thermal comfort and reduces energy consumption.

H3. The enhancement effects of XGBoost combined with Deep Q Network (XGB-DQN) on the strategies of HVAC-window systems for comfort and energy efficiency differ across various climate zones.

3. Material and methods

The construction process of the XGB-DQN algorithm primarily involves four modules: (1) data preprocessing, (2) XGBoost construction (including feature engineering, hyper-parameter tuning, and model training), (3) Deep Q Network model construction (defining states, actions, reward functions, etc.), and (4) assembling XGBoost and DQN into an iterative and interactive program. The specific procedural framework is illustrated in Fig. 2, and subsequent sections delve into the algorithm's principles, deployment and evaluation processes.

3.1. Data source and processing

The data for this study is sourced from the ASHRAE Global Building Occupant Behavior Database, contributed by researchers from 15 countries and 39 institutions, covering 10 distinct climate zones (<https://ashraeobdatabase.com/#/>) [9,44]. To enhance the usability and accuracy of the data, several preprocessing steps are employed in this study: (1) filtering of features and selecting datasets containing the required variables; (2) standardization of the time step across all datasets to 1 h; (3) removal of missing values and correction of data records conflicting with the original dataset; (4) expansion and supplementation of features through the process of feature engineering.

3.1.1. Selection of datasets

The ASHRAE Global Building Occupant Behavior Database aims to enhance knowledge and comprehension of realistic occupancy patterns and human-building interactions with building systems [45]. This database comprises long-term time-series data obtained through on-site experimental measurements, providing a valuable empirical foundation for studying building performance.

For this study, five datasets are carefully selected to explore the impact of window and HVAC system operations on indoor environments. Two key criteria are defined during the data set selection process: firstly, the chosen datasets must include four crucial variables—indoor environment, outdoor environment, HVAC operation, and window operation (Table 1); secondly, indoor environmental variables should encompass at least two key parameters, namely temperature and relative humidity.

Ultimately, datasets 6, 10, 16, 21, and 23 are selected out, encompassing records from a total of 18 rooms across four experimental locations: Sao Paulo and Florianopolis in Brazil, Nanjing in China, and Rende in Italy. These four cities are situated in different climatic regions: Sao Paulo, Florianopolis, and Nanjing fall under the Cfa category (humid subtropical climate), while Rende belongs to the Csa category (hot-summer Mediterranean climate). Details, including the data characteristics of the database, the temporal coverage of available datasets, the climatic characteristics of the four cities, the annual outdoor temperature trends, and corresponding schematic diagrams of adaptive thermal comfort ranges, are explicitly presented in Appendix B.

3.1.2. Feature engineering

To improve the predictive performance of machine learning models, it is essential to extend the dimensions of the original data set through feature engineering (Table 2). This process aims to capture hidden information effectively. Feature engineering primarily employed two methods: one involved further statistical calculations on the raw data, such as extracting month and hour information from timestamps and computing the operating duration of each device within each time step based on the states of the air conditioning and windows. Combining the cumulative operating duration recorded in the preceding time steps allowed the estimation of the total operating time for each device.

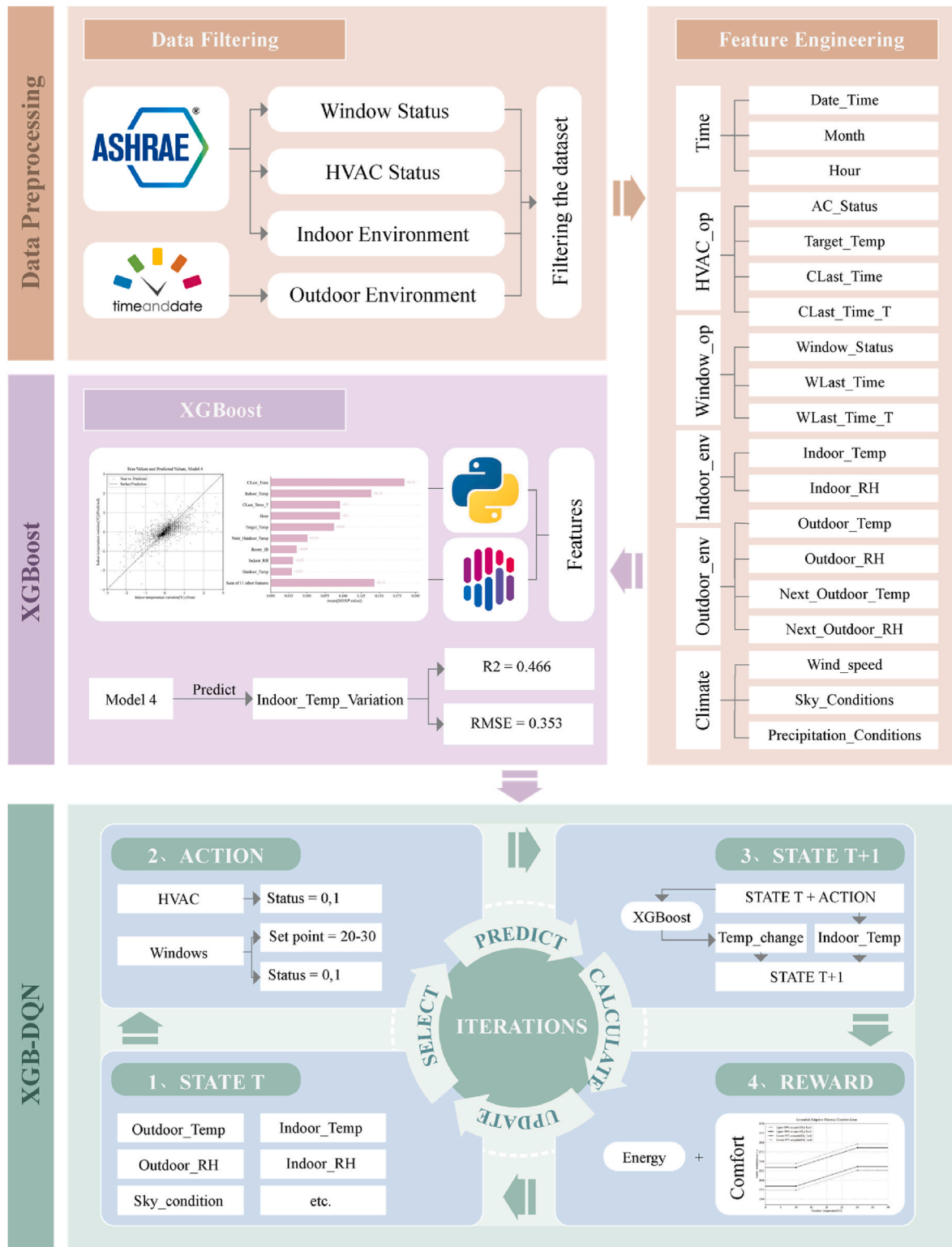


Fig. 2. Research framework.

Additionally, to address missing and anomalous outdoor environmental data in some datasets, this study utilized historical weather data from Time and Date (<https://www.timeanddate.com/>) to impute information such as outdoor temperature, humidity, wind speed, sky conditions, and precipitation conditions. After removing rows containing missing values, text-type data and numerical data were mapped using label encoding. In the end, a total of 42,952 records and 20 feature variables

were obtained.

3.2. Synergistic control strategies: integrating XGB and DQN

The Synergistic Control Strategies introduced in this study propose an integrated control approach that combines DQN and XGBoost. XGBoost is employed to predict the change in indoor temperature

Table 1
ASHRAE occupant behavior data-set information [9].

Dataset ID	Country	City	Climate Type	Outdoor Env.	Indoor Env.	HVAC Status	Window Status
6	Brazil	Florianopolis	Cfa	×	×	×	×
25 [46]	Brazil	Florianopolis	Cfa	×	×	×	×
16	Brazil	Sao Paulo	Cfa	×	×	×	×
23 [47]	Brazil	Sao Paulo	Cfa	✓	×	×	×
21	China	Nanjing	Cfa	×	×	×	×
15	China	Shanghai etc.	Cfa	×	×	×	×
10 [48]	Italy	Rende	Csa	✓	×	×	×
4 [49]	Denmark	Odense	Cfb	✓	×	×	×
7 [50]	Australia	Bangholme	Cfb	×	×	×	×
17	China	Shanghai	Cfa	×	×	×	×
8 [51]	Canada	Toronto	Dfb	×	×	×	×
14	China	Beijing	Dwa	×	×	×	×
5 [52]	Italy	Perugia	Csa	×	×	×	×
20 [53]	Austria	Vienna	Cfb	×	×	×	×
24 [54]	Germany	Frankfurt	Cfb	✓	×	×	×
18 [55]	UAE	Abu Dhabi	Bwh	✓	×	×	×
30 [56]	USA	Austin	Cfa	✓	×	×	×
32 [57]	USA	Berkeley	Csb	✓	×	×	×
33	USA	Unknown	Unk.	✓	×	×	×
2 [58]	USA	Lincoln	Dfa	✓	×	×	×
9 [59]	Canada	Ottawa	Dfb	✓	×	×	×
11 [60]	USA	San Antonio	Cfa	✓	×	×	×
22 [61]	USA	Palo Alto	Csb	✓	×	×	×
13 [62]	Poland	Poznan	Cfb	×	×	×	×

Note: Symbol × represents the presence of this type of data, and ✓ indicates that data can be supplemented through public records.

Table 2
Features and description.

Categories	Features	Description	Source
Time	Variation_Indoor_Temp	Variation of indoor temperature after an operation	Derived data
	Date_Time	Date and time of the recorded data [Y/m/d H:M:S]	Raw data
	Month	Month when data was recorded [month]	Derived data
	Hour	Hour when data was recorded [Hour]	Derived data
AC operations	AC_Status	State of Air-conditioner [0 = off, 1 = on]	Raw data
	Target_Temp	Target temperature set of the air conditioning [°C]	Derived data
	CLast_Time	Last time the air conditioner was activated [min]	Derived data
	CLast_Time_T	Duration of the last air conditioner activation [min]	Derived data
Window operations	Window_Status	State of Window [0 = off, 1 = on]	Raw data
	WLast_Time	Last time the window was opened [min]	Derived data
	WLast_Time_T	Duration of the last window openness [min]	Derived data
	Indoor_Temp	Current indoor temperature [°C]	Raw data
Indoor environment	Indoor_RH	Current indoor relative humidity [%]	Raw data
	Outdoor_Temp	Current outdoor temperature [°C]	Web crawler data
	Outdoor_RH	Current outdoor relative humidity [%]	Web crawler data
	Next_Outdoor_Temp	Outdoor temperature for the next time step [°C]	Web crawler data
Outdoor environment	Next_Outdoor_RH	Outdoor relative humidity for the next time step [%]	Web crawler data
	Wind_speed	Current wind speed [km/h]	Web crawler data
	Sky_Conditions	Description of the atmospheric conditions [Passing cloud, Sunny, Partly cloudy, Fog et al.]	Web crawler data
	Precipitation_Conditions	Description of precipitation conditions [Light rain, Drizzle, Thundershowers, Rain et al.]	Web crawler data

resulting from the implementation of a specific HVAC-window behavior. In parallel, DQN is utilized to construct a feedback network assessing the relationship between the environment and behaviors. This network facilitates the selection of the optimal strategy for the HVAC-window system to adjust indoor temperature. The algorithmic principles and specific deployment of DQN and XGBoost will be elaborated upon in the following sections.

3.2.1. XGBoost-driven state prediction

XGBoost is an extreme gradient boosting algorithm based on decision tree [63]. It iteratively constructs multiple decision trees (Equation (1)) to continually fit residuals, ultimately building a predictive model applicable to regression or classification. Assuming the presence of K trees, the predicted value $\hat{y}_i$ for the i-th sample is obtained through the ensemble of K trees.

$$\hat{y}_i = \sum_{k=1}^K f_k(x_i) \quad (1)$$

Gradient Boosting entails an iterative process wherein, at each iteration, the gradient descent algorithm is employed to identify a decision tree (weak learner) aimed at minimizing the error between predicted and true values (Equation (2)). The objective function $\mathcal{L}(\varphi)$ is articulated as follows, where $l(y_i, \hat{y}_i)$ denotes the loss function, typically a convex function utilized to characterize the disparity between predicted value $\hat{y}_i$ and actual value y_i . The parameter $\Omega(f) = \gamma T + \frac{1}{2}\lambda \|\omega\|^2$ represents the regularization term, controlling the complexity of the model to mitigate the risk of over-fitting and enhance the model's generalization capability. Therefore, upon completion of iterations, multiple weak learners combine to form a robust ensemble model.

$$\mathcal{L}(\varphi) = \sum_{i=1}^n l(y_i, \hat{y}_i^{(t-1)} + f_t(x_i)) + \Omega(f_t) \quad (2)$$

In addition to introducing regularization terms into the loss function, it allows for a second-order Taylor expansion of the objective function (Formula 3), enhancing computational precision. Here, $l(y_i, \hat{y}_i^{(t-1)})$ represents the constants known from the previous iteration, with these constants being independent of function optimization and thus typically omitted for simplification. In Equation (3), $g_i = \partial_{y_i^{(t-1)}} l(y_i, \hat{y}_i^{(t-1)})$ represents the first-order derivative of $f_i(x_i)$, while $h_i = \partial_{y_i^{(t-1)}}^2 l(y_i, \hat{y}_i^{(t-1)})$ signifies the second-order derivative. Furthermore, XGBoost employs an Approximate Greedy Algorithm to select leaf nodes with the maximum gain after splitting, bypassing the need to traverse all splitting points and consequently improving training speed. Grid search, Random search, and Bayesian optimization are commonly employed methods for tuning parameters in the XGBoost algorithm [64]. Bayesian optimization, based on observed prior knowledge, facilitates subsequent optimization and is capable of identifying globally optimal parameters. Consequently, Bayesian optimization is selected to fine-tune hyper-parameters in the XGBoost algorithm.

$$\mathcal{L}(\varphi) \cong \sum_{i=1}^n \left[l\left(y_i, \hat{y}_i^{(t-1)} + g_i f_i(x_i) + \frac{1}{2} h_i f_i^2(x_i)\right) \right] + \Omega(f_i) \quad (3)$$

This study utilizes the XGBoost method to predict indoor temperature variations under different combinations of HVAC and window operations. A total of 17 features are chosen as independent variables, comprising 9 features such as indoor and outdoor temperatures, humidity, and 7 features including air conditioning and window states, as well as operating time, along with 3 features such as month, hour, and room identifier. The data set is split into training and testing sets with an 80 %–20 % ratio. Employing the Bayesian algorithm, the selected hyper-parameters include a decision tree depth (max_depth) of 5, a learning rate (learning_rate) of 0.27, and an estimator count (n_estimators) of 500.

3.2.2. DQN-driven strategy generation

RL involves constructing a core framework where the pivotal element is the creation of an environment for an intelligent agent. This agent continually experiments with various action strategies, interacts with the environment, and generates feedback. Throughout this iterative process, the primary components encompass State_t, Action_t, Reward_t, State_{t+1}. Here, State_t represents the state at time t , Action_t is the action strategy taken from the action space, State_{t+1} is the subsequent state after applying Action_t to State_t, and the environment then provides a reward, Reward_t, based on State_{t+1}. The agent identifies actions that lead to superior outcomes, gradually refining its strategy to maximize cumulative rewards or achieve specific objectives.

Addressing how the environment computes State_{t+1} based on State_t and Action_t, RL can be broadly classified into two major categories: Model-Free and Model-Based RL. Model-Free algorithms directly acquire empirical data through the interaction between the agent and the real environment. Due to their ability to operate without prior knowledge of the environment model, they are applicable to a variety of problems, especially those with dynamic or difficult-to-model environments. Conversely, Model-Based reinforcement learning relies on an environment model, which includes state transition probabilities and reward functions. This model can be inferred through methods such as software simulation or physical modeling. Given the assurance of model quality and accuracy, Model-Based approaches may be more efficient than Model-Free methods. In this study, employing the XGBoost model to predict State_{t+1} eliminates reliance on an environment model, making it more suitable for Model-Free reinforcement learning methods.

The selection of reinforcement learning methods is also dependent on the nature of the problem and the characteristics of the environment, as different methods have corresponding areas of applicability. Q-Learning is one of the most classic methods, centered on learning a Q-value function, $Q(s, a)$, to evaluate long-term rewards for each possible state-action combination. Consequently, it is more suitable for problems

with finite states and actions. However, in practical indoor environments, the indicators of environmental states often exhibit continuous changes, making the Q-table unable to cover all possible combinations of states and actions. In such cases, Deep Q Network (DQN), a Model-Free RL method that combines Deep Neural Networks with Q-learning to estimate value functions, is selected as the primary research approach due to its superior ability to handle continuous and complex state spaces. The DQN settings employed in this study are outlined as follows.

● State Space:

The state represents the information obtained by the agent from the environment at a specific moment. In this research, the state parameters include outdoor weather parameters such as outdoor temperature, outdoor humidity, sky conditions, precipitation conditions, wind speed, indoor parameters like indoor temperature and humidity, as well as time. These parameters collectively form the state space S_t in Equation (4):

$$S_t = \left\{ \begin{array}{l} \text{Hour, Wind_speed, Sky_} \\ \text{Conditions, Precipitation_} \\ \text{Conditions,} \\ \text{Indoor_Temp, Indoor_RH, Outdoor_Temp, Outdoor_RH} \end{array} \right\} \quad (4)$$

● Action Space:

The control of actions is primarily reflected in two aspects. One is the user's manipulation of the window $A_{\text{Window}} = \{0, 1\}$, with 0 indicating closed and 1 indicating open. The other is the user's control of the air conditioning unit $A_{\text{AC}} = \{0, 1\}$, with 0 indicating off and 1 indicating on. Drawing from user behaviors in adjusting the air conditioner temperature via a remote control, the action space incorporates the value of the target temperature $A_{\text{Target temp}} = \{20, 21, 22, \dots, 30\}$. Therefore, the action space is represented as $A_t = A_{\text{Window}} \times A_{\text{AC}} \times A_{\text{Target}}$, and the correspondence between indices and actions is shown in Table 3.

● Reward Function:

Upon the execution of specific actions by the agent, the environment undergoes changes, and the modified environment yields rewards for the intelligent agent. Appendix C summarizes the approaches used in prior research for establishing reward functions. In this study, the reward function, denoted as R_t (Formula 5), comprehensively considers three aspects: the occupant's usage pattern β , indoor air temperature comfort R_{comfort} , and HVAC system energy consumption R_{Energy} .

Firstly, drawing from Sun et al.'s research, it is observed that the predominant period of occupancy for office users is concentrated between 7:00 a.m. and 7:00 p.m. [65]. Consequently, the XGB-DQN system operates from 6:00 a.m. to 6:00 p.m., with a corresponding control policy set each hour. To ensure the accuracy of the system's reward

Table 3

Mapping table for action space and encoding.

Code	A_{Window}	A_{AC}	$A_{\text{Target temp}}$	Code	A_{Window}	A_{AC}	$A_{\text{Target temp}}$
0	0	0	\	12	1	0	\
1	0	1	20	13	1	1	20
2	0	1	21	14	1	1	21
3	0	1	22	15	1	1	22
4	0	1	23	16	1	1	23
5	0	1	24	17	1	1	24
6	0	1	25	18	1	1	25
7	0	1	26	19	1	1	26
8	0	1	27	20	1	1	27
9	0	1	28	21	1	1	28
10	0	1	29	22	1	1	29
11	0	1	30	23	1	1	30

Table 4

Model training and testing results.

	Model 1		Model 2		Model 3		Model 4	
	Training set	Test set	Training set	Test set	Training set	Test set	Training set	Test set
MAE	0.2168	0.2358	0.1423	0.1850	0.2118	0.2327	0.1322	0.1827
RMSE	0.3910	0.4529	0.2462	0.3648	0.3834	0.4581	0.2290	0.3526
R ²	0.9800	0.9725	0.9921	0.9821	0.3881	0.1985	0.8127	0.4664

function in reflecting real-world usage scenarios, a decision coefficient β is introduced to measure whether it is working hours. When the time is between 6:00 a.m. and 6:00 p.m., the decision coefficient β is set to 1; otherwise, it is set to 0 [66].

The core components of the reward function R_t comprise the indoor thermal comfort reward R_{comfort} and HVAC system energy-saving reward R_{Energy} . In this context, weights ω_1 and ω_2 are utilized to fine-tune the relative contributions of thermal comfort and energy consumption to the overall reward (Equation (5)). A larger value for ω_1 suggests a heightened inclination of the intelligent agent for optimizing thermal comfort, while a smaller ω_1 implies a bias towards energy savings [66]. In this study, thermal comfort and energy consumption are considered to have equal significance, leading to the setting of ω_1 and ω_2 as 1 [33].

Within R_{comfort} , if the indoor temperature falls within the comfort boundary (Equation (8)), the reward baseline for indoor thermal comfort is set to 0 [4]. When the indoor temperature deviates from the thermal comfort zone, the reward function imposes a penalty by subtracting the square of the temperature difference between indoor temperature and the comfort boundary (Equation (6)) [6]. R_{Energy} represents the penalty term for energy consumption intensity, measured by the product of air conditioning operation time t_{HVAC} and rated power P (Equation (7)) [67]. R_{Energy} is 0 when the air conditioning is off and equals $t_{\text{HVAC}} \times P$ when the air conditioning is on with closed windows. In the event of open windows during air conditioning operation, considering increased energy waste, the penalty value is doubled. Recognizing the substantial unit differences between R_{comfort} and R_{Energy} , an amplification factor λ is introduced to balance the dimensions [33].

$$R_t = \beta(\omega_1 R_{\text{comfort}} + \omega_2 \lambda R_{\text{Energy}}) \quad (5)$$

$$R_{\text{comfort}} = \begin{cases} 0, & \text{if } T_{\text{in}} \in [T_{\text{lower}}, T_{\text{upper}}] \\ -\min(|T_{\text{in}} - T_{\text{upper}}|, |T_{\text{in}} - T_{\text{lower}}|)^2, & \text{if } T_{\text{in}} \notin [T_{\text{lower}}, T_{\text{upper}}] \end{cases} \quad (6)$$

$$R_{\text{Energy}} = \begin{cases} 0, & A_{\text{AC}} = 0 \\ -t_{\text{HVAC}} \times P, & A_{\text{AC}} = 1 \text{ and } A_{\text{Window}} = 0 \\ -2 \times t_{\text{HVAC}} \times P, & A_{\text{AC}} = 1 \text{ and } A_{\text{Window}} = 1 \end{cases} \quad (7)$$

$$[T_{\text{lower}}, T_{\text{upper}}] = \begin{cases} [18.4, 23.4], & t_{\text{out}} \leq 10^\circ\text{C} \\ [15.1 + 0.31t_{\text{out}}, 20.1 + 0.31t_{\text{out}}], & 10^\circ\text{C} < t_{\text{out}} < 30^\circ\text{C} \\ [24.5, 29.6], & t_{\text{out}} \geq 30^\circ\text{C} \end{cases} \quad (8)$$

The agent, guided by environment-derived rewards, continually adjusts its strategy to maximize the cumulative sum of discounted future rewards (Equation (9)). In RL, the importance of future rewards may not align with that of current rewards, with earlier rewards often carrying more significance than later ones. Consequently, a discount factor γ is introduced to balance the relative importance of current and future rewards. A higher γ emphasizes long-term rewards, while a lower γ focuses more on immediate rewards. DQN employs two neural networks to approximate Q-values, denoted as $Q(s, a)$. The optimal Q-value is estimated and updated using the Bellman Optimality Equation (Equation (10)), allowing the agent to gradually learn the optimal strategy for taking different actions in different states. The learning rate α determines how much weight new observational data should carry during each Q-value update, contributing to a balance between stability and convergence speed in the learning process.

$$R_t = \sum_{i=0}^T \gamma^i R_{t+i+1} \quad (9)$$

$$Q_{\text{new}}(s_t, a_t) = Q_{\text{old}}(s_t, a_t) + \alpha \times \left[r_t + \gamma \times \max_a Q(s_{t+1}, a) - Q_{\text{old}}(s_t, a_t) \right] \quad (10)$$

3.2.3. Integration of XGB and DQN

This study introduces an HVAC-window system collaborative control framework (Fig. 3) that ingeniously integrates XGB-DQN. This framework is applicable in virtually any real-world building environment as long as the variations in the environment are known. Features marked with solid circles represent state and action parameters that require on-site measurements at hourly intervals. For instance, indoor temperature and humidity can be measured using temperature sensors, while the on/off status of air conditioning and windows can be controlled through signal sensors. In contrast, features marked with hollow circles do not require direct measurements and can be deduced from measured features or completed using web crawling. Once feature processing is complete, these features are input into the XGBoost model for training. The model outputs temperature changes for different HVAC-window strategies and predicts the indoor temperature for the next hour.

Fig. 3 illustrates the operational decision-making process of the HVAC-window system in a specific room at a given moment. Initially, the environmental state S_t is acquired through sensors (Process 1). This state serves as the initial condition, transmitted to a neural network within the RL framework (Process 2). Subsequently, the agent selects a set of actions A_t (Process 3), and the mapping of these actions, integrated with the data from the initial state, constructs a new data set that is fed into XGBoost (Process 4). The model outputs the indoor environment predictions for the next moment (Process 5). Based on the predictions, the next state S_{t+1} is updated and recorded (Process 6). The thermal comfort score for the indoor temperature and energy consumption penalty score at time S_{t+1} are computed and recorded as R_t (Process 7). The iteration results $\{S_t, A_t, R_t, S_{t+1}\}$ are obtained (Process 8) and stored in memory for sampling training of the neural network (Process 9). Finally, as the neural network converges to stability, a continuous HVAC-window control strategy that balances energy consumption and indoor thermal comfort can be obtained by inputting an initial state.

3.3. Evaluation

3.3.1. Accuracy evaluation of XGB

The key to ensuring the XGB-DQN program generates a reasonable optimal strategy lies in accurately predicting the indoor environmental changes at the next time step through XGB. However, there are performance differences among XGB models constructed using different modeling methods. To select the model that most accurately captures indoor temperature changes, this study constructs and evaluates four different XGB models representing different prediction targets and data processing methods during the testing phase: Model 1 directly predicts indoor temperature values using raw data; Model 2 directly predicts indoor temperature values using data extended through feature engineering; Model 3 predicts the change in indoor temperature using raw data, indirectly obtaining indoor temperature values; Model 4 predicts the change in indoor temperature using feature-engineered data,

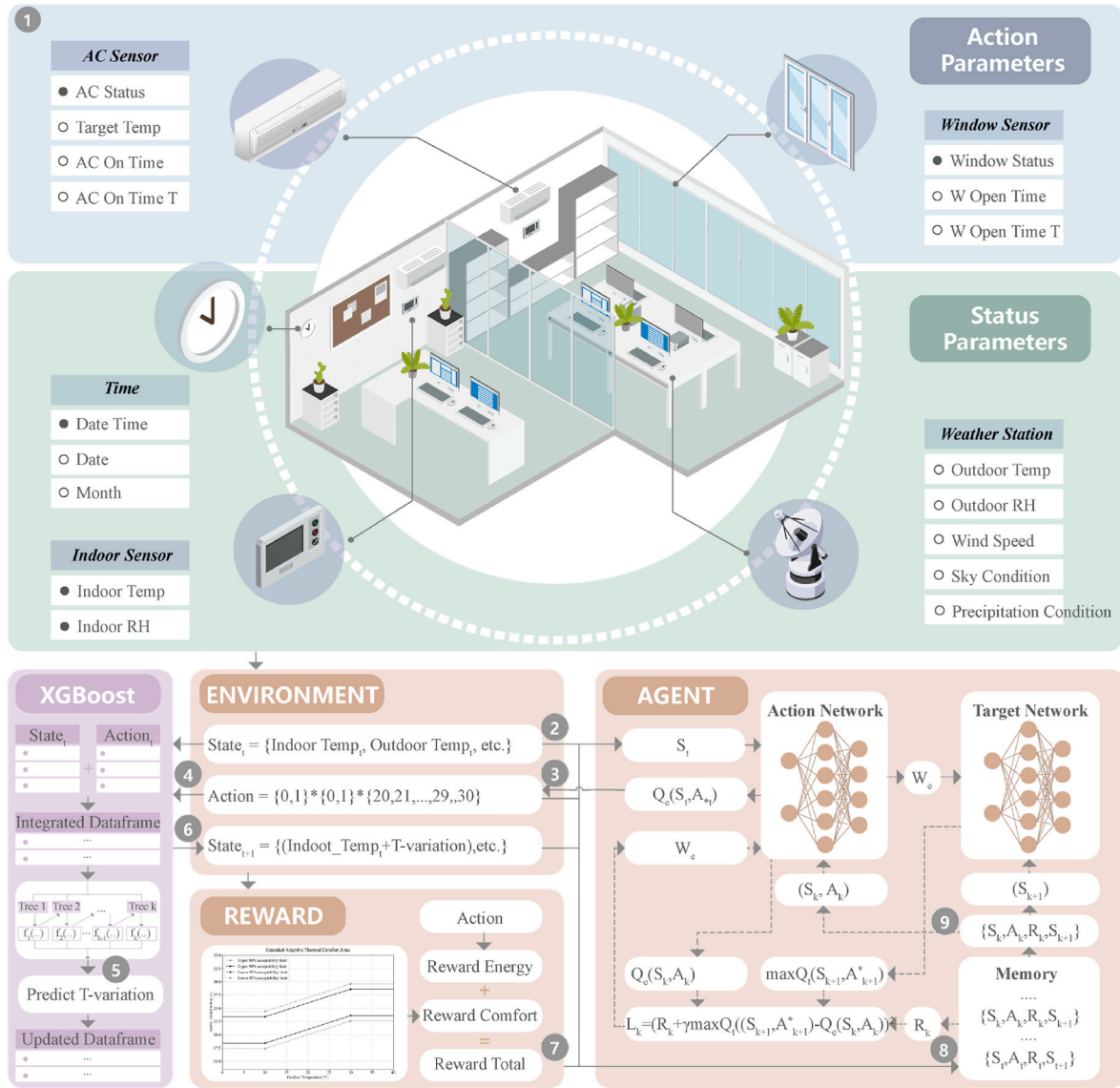


Fig. 3. Program structure of XGB-DQN for HVAC-Window system control.

indirectly obtaining indoor temperature values.

Given that predicting indoor temperature changes involves a regression model with continuous variables, this study employs three metrics to evaluate the model's predictive performance (Equation (4)): (1) Mean Absolute Error (MAE), the average absolute difference between actual observations and model predictions; (2) Root Mean Square Error (RMSE), the root mean square difference between actual observations and model predictions; (3) and R-squared (R^2): the percentage of variance in actual observations explained by the model. Generally, smaller MAE and RMSE values and a larger R^2 signify a better fit and higher predictive accuracy of the model to the actual data.

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|, RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}, R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y}_i)^2} \quad (11)$$

3.3.2. Convergence evaluation of XGB-DQN

The assessment of reinforcement learning model performance typically relies on the analysis of Reward Curve [33]. Reward Curve typically plots training iteration numbers on the x-axis and cumulative

rewards on the y-axis, where cumulative rewards represent the total rewards accumulated by the agent during interaction with the environment. The shape and trend of this curve reflect whether the model is gradually stabilizing. Two methods can be employed for calculating cumulative rewards: one involves recording agent rewards within a certain number of steps and averaging them, while the other entails averaging the Q-values of a certain quantity of state-action pairs [37]. Considering the potential impact of initial states and environmental randomness on reinforcement learning algorithms, conducting multiple experiments can effectively mitigate the influence of these random factors, providing a more robust performance assessment. Therefore, in this study, a sample room is randomly selected in four cities, and the mean rewards for all initial state scenarios are calculated during different training episodes. The trend of these changes is plotted to assess the model's performance when facing different environments and initial states.

3.3.3. Performance optimization of XGB-DQN

This research conducts a comparative evaluation of the application potential of the XGB-DQN model by constructing three distinct HVAC-window operation scenarios: (1) Baseline model, portraying the natural state without HVAC-window operations; (2) Occupant control,

reflecting spontaneous HVAC-window operations from the original data set; and (3) XGB-DQN, representing an optimized HVAC-window operation strategy through reinforcement learning algorithms. The baseline model serves as a reference point to gauge whether other models can significantly surpass baseline performance. Performance metrics utilized in the study encompass thermal comfort, energy efficiency (absolute and relative energy consumption), and electricity cost, with the aim of quantifying and evaluating the potential of XGB-DQN in enhancing thermal comfort levels, reducing energy consumption, and lowering electricity expenses.

Thermal comfort performance is quantified as the percentage increase in the duration of thermal comfort compared to the baseline model following the implementation on a specific model (Equation (12)). Energy efficiency comprises absolute and relative energy consumption, with absolute energy consumption representing the total additional energy usage incurred when using a specific model or system compared to the baseline model (Equation (13)). Relative energy consumption considers the energy required to improve thermal comfort per unit time, thereby controlling the endogenous effects caused by the duration of thermal comfort [68] (Equation (14)).

The estimation of electricity costs refers to the projected electricity expenditure resulting from the operation of air conditioning equipment, either spontaneously by users or under the control of the XGB-DQN algorithm, involving parameters such as standard electricity prices, the operating time of air conditioning equipment, and the rated power of split air conditioners. As specific information on air conditioner power and electricity rates is not available in the original data set, the study employs a general estimation method (Equation (15)). Initially, standard electricity rate data are obtained from the Global Petrol Prices official website (<https://www.globalpetrolprices.com/>). As of June 2023, commercial electricity rates in China, Brazil, and Italy are \$0.077 (0.634 CNY), \$0.147 (0.712 BRL), and \$0.591 (0.535 EUR), respectively. Additionally, the rated power of the air conditioner is estimated based on empirical values from previous relevant studies, with estimated powers for split air conditioners in sample rooms in four cities (Nanjing, Sao Paulo, Florianopolis, and Rende) being 0.72 kW [69], 0.86 kW, 0.86 kW [70], and 1.5 kW [71], respectively. Finally, to standardize the temporal dimension, a time conversion factor is multiplied to assess the annual electricity cost.

$$\text{Thermal comfort} = \frac{H'_{\text{comfort}} - H_{\text{comfort}}}{H_{\text{total}}} \times 100\% \quad (12)$$

$$\text{Absolute Energy Consumption} = P_{\text{rated}} \times (H'_{\text{operation}} - H_{\text{operation}}) \quad (13)$$

$$\text{Relative Energy Consumption} = \frac{P \times (H'_{\text{operation}} - H_{\text{operation}})}{H'_{\text{comfort}} - H_{\text{comfort}}} \quad (14)$$

$$\begin{aligned} \text{Estimated electricity cost} &= P_{\text{rated}} \times (H'_{\text{operation}} - H_{\text{operation}}) \times E_{\text{price}} \\ &\times \frac{(365 \times 24)}{H_{\text{total}}} \end{aligned} \quad (15)$$

3.3.4. Weekly case analysis

To facilitate a cleaner and more straightforward comparison of the XGB-DQN model's performance relative to occupant spontaneous control and the baseline model over a specific time frame, this study employs detailed data from each city as individual cases. The selected data adheres to the following criteria: (1) data records reflecting continuous 24-h temporal changes, (2) data records spanning seven consecutive natural days, (3) cases where the rooms were occupied by occupants on the majority of days, and (4) data reflecting indoor and outdoor environments indicative of local climate characteristics. Ultimately, data from Florianopolis for February 15th to 21st, 2020, Nanjing for September 6th to 12th, 2016, Sao Paulo for October 3rd to 9th, 2017,

and Rende for July 25th to 31st, 2016, are selected as examples.

4. Results

4.1. Performance evaluation of XGB-DQN

4.1.1. Accuracy evaluation of XGB

The reliable operation of XGB-DQN hinges on the accurate predictions of the XGB model. A comprehensive testing and evaluation are conducted for four different modeling approaches. The results indicate a significant variation in the accuracy of the XGB model under different settings. Models 1 and 2 employed a direct prediction method for the indoor temperature at the next time step using features from the current time step. During the training process, Models 1 and 2 achieved R^2 values of 0.97 and 0.98, respectively, approaching 1, which typically signifies excellent model fitting. However, when these models predict the temperature trend for a future period based on a given initial state, they deviate significantly from the ground truth (Fig. 4). This discrepancy arises due to the strong substantial influence of past values of indoor temperature data on the current value, a phenomenon commonly referred to as Temporal Auto-regression or Auto-correlation [72], resulting in inflated R^2 statistical outcomes.

To eliminate the interference of auto-correlation, models 3 and 4 are introduced, utilizing indoor temperature changes as predictive variables to indirectly predict the indoor temperature at the next time step. Through training, both MAE and RMSE statistics exhibited significant reductions, and over an extended period, the predicted results closely aligned with the actual temperature trends. In comparison to model 3, model 4 underwent additional feature engineering and data cleaning, resulting in superior performance (R^2 increased from 0.199 to 0.466, RMSE decreased from 0.458 to 0.353). Simultaneously, there is a notable enhancement in the data fitting trend (Fig. 4). Ultimately, model 4 demonstrated the highest logical consistency and prediction accuracy, earning its selection as the predictive model for state iteration in the HVAC-window collaborative control system. (see Table 4).

4.1.2. Convergence evaluation of XGB-DQN

The shape and trend of the reward curve serve as an intuitive reflection of whether the model is gradually achieving stability (Fig. 5). In the initial training phase of the XGB-DQN model, the average cumulative reward is relatively low due to the random selection of action policies. However, with continuous trial and error and the accumulation of experience, XGB-DQN gradually learns, and the cumulative reward for each episode steadily increases. Eventually, the XGB-DQN exhibits a stable convergence trend in the reward curves across sample rooms in all cities.

4.2. Performance optimization of XGB-DQN

4.2.1. Indoor thermal comfort

This study conducted a comparative analysis of indoor thermal comfort performance under three scenarios during working hours. The horizontal axis of the graph represents outdoor temperature, while the vertical axis represents indoor temperature (Fig. 6). The heat-map employs circle size to denote the frequency of occurrence for a particular indoor-outdoor state, with color indicating the probability of activating the air conditioning system in that state. Overall, the XGB-DQN effectively adjusts HVAC-window states to shift the uncomfortable indoor environment in the baseline model into the thermal comfort range, outperforming occupant-controlled adjustments (Table 5). In comparison to the baseline model, the duration of indoor thermal comfort increased from 12,768 h to 17,222 h (24 % increase) under XGB-DQN control.

More specifically, there are variations in the enhancement of indoor thermal comfort provided by the XGB-DQN model across room samples in different cities. According to the numerical results of the Thermal

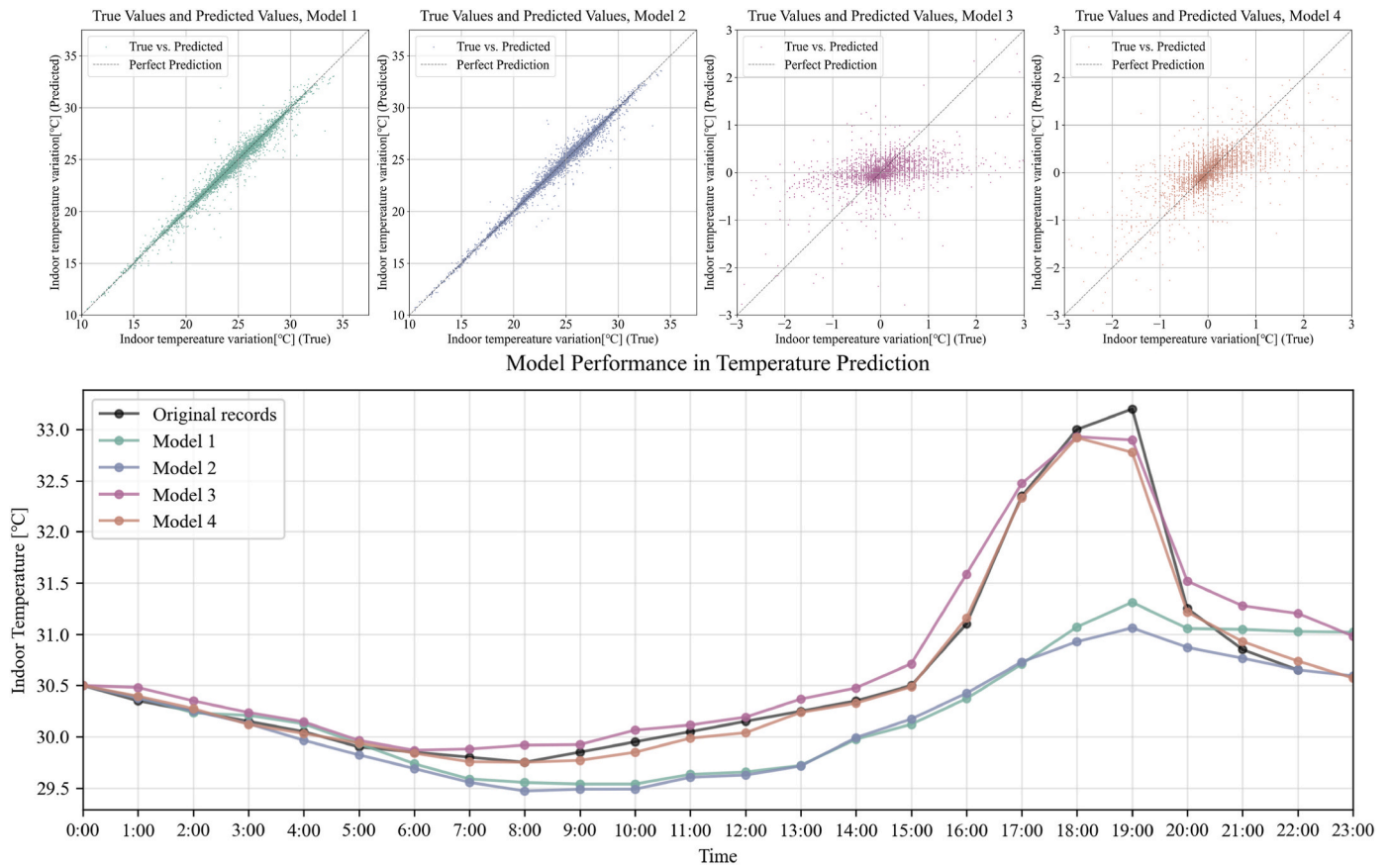


Fig. 4. Fitting performance of four XGBoost prediction models.

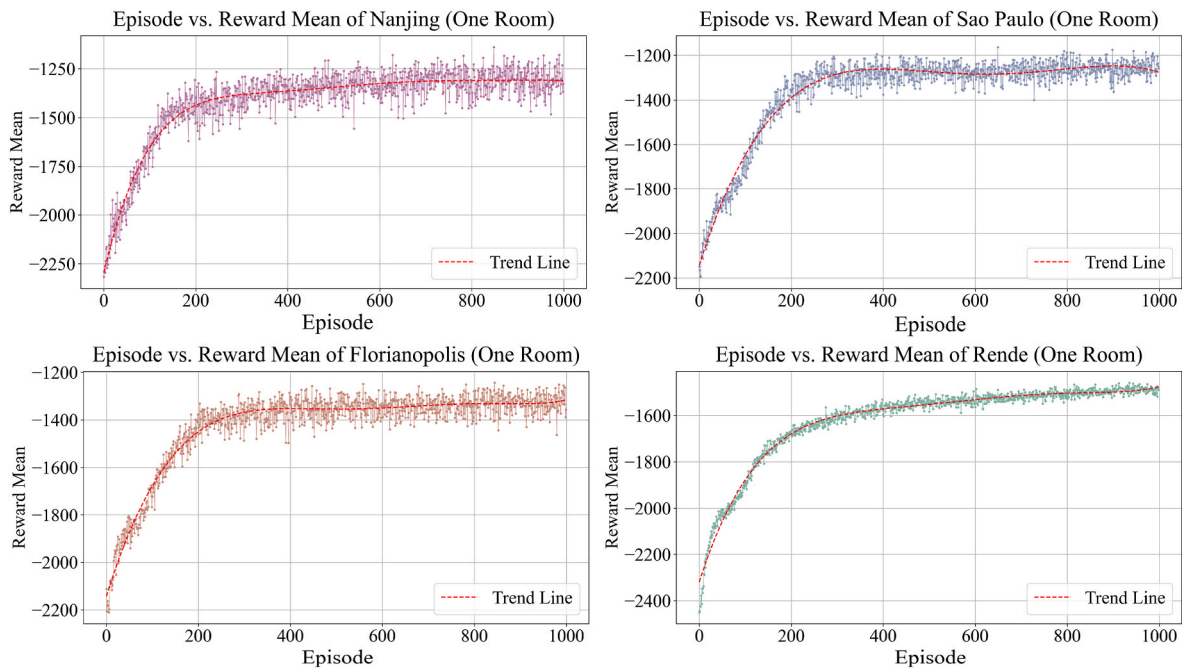


Fig. 5. XGB-DQN reward curve variations across four urban room samples.

Comfort indicator, tested room in Rende show the highest potential, demonstrating a 36 % improvement in thermal comfort compared to the baseline model. Following closely are room samples from Florianopolis and Sao Paulo, both located on the southeastern coast of Brazil, with

similar enhancement effects of 23 % and 21 %, respectively. This similarity may be attributed to their comparable geographical locations, climate characteristics, and building features. Lastly, room samples in Nanjing, China, show an 11 % improvement in thermal comfort with

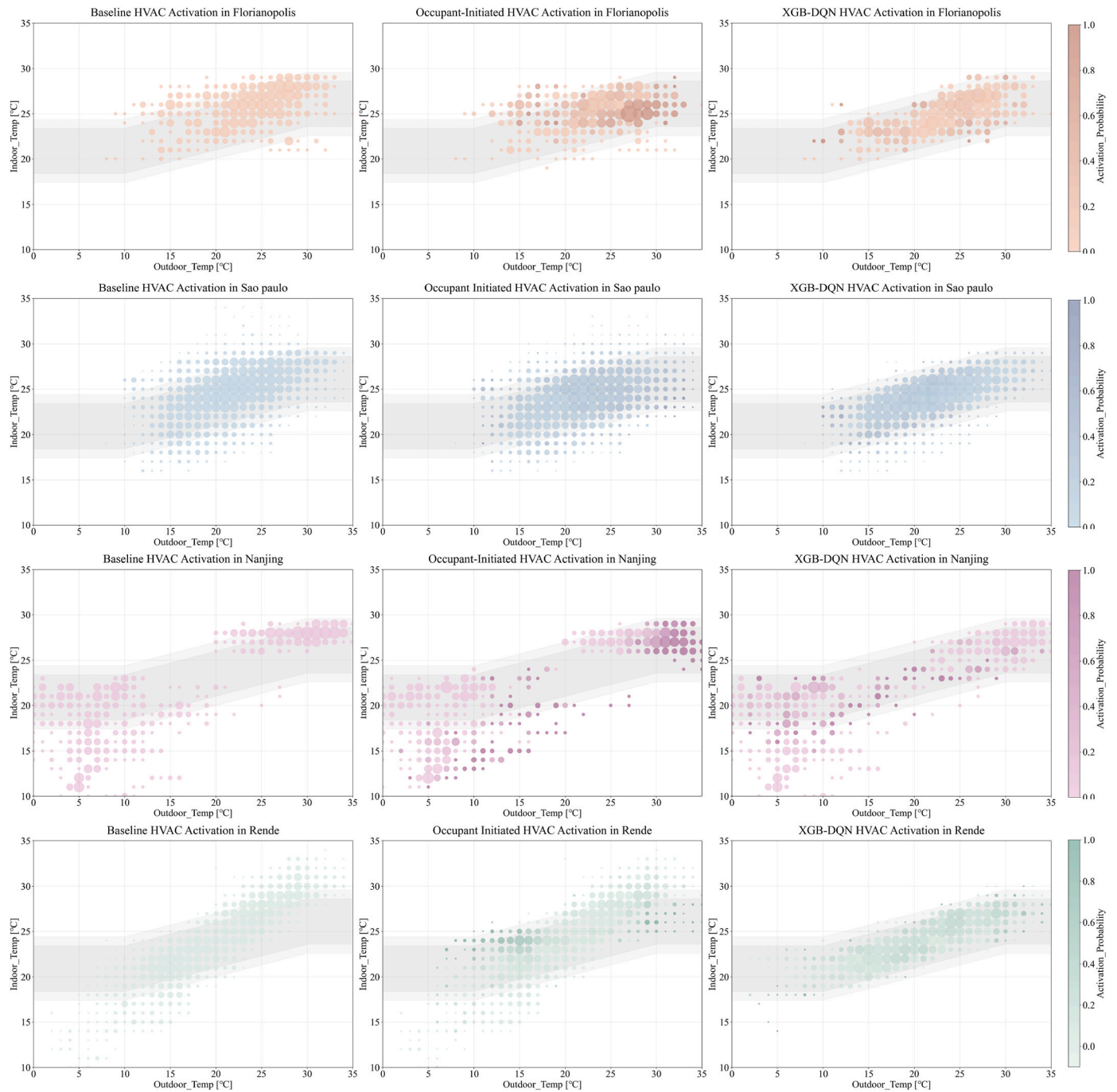


Fig. 6. Temperature distribution, along with HVAC activation probability heatmaps for room samples in four cities under three models.

XGB-DQN compared to the baseline model.

4.2.2. Energy consumption and electricity cost

XGB-DQN enhances indoor comfort and simultaneously achieves a significant reduction in energy consumption. Absolute Energy Consumption reveal that under the control strategy of XGB-DQN, the overall HVAC runtime for room samples in three cities, excluding Rende, decreases by 780 h (32 %). This reduction represents considerable energy savings. In the case of Rende's room samples, despite a 9.6 % increase in air conditioning runtime (85 h), the improvement in indoor thermal comfort reached the highest proportion at 36 %. Therefore, relative Energy Consumption is introduced to further evaluate the energy required for the enhancement of unitary thermal comfort time improvement. Statistical results demonstrate a notable reduction in

relative energy consumption for all city room samples. Particularly, Florianopolis experiences the most significant decline (from 0.32 to 0.09), demonstrating XGB-DQN's ability to minimize unnecessary air conditioning energy consumption while ensuring thermal comfort.

Another crucial metric is the estimated annual electricity cost. In terms of economic performance, XGB-DQN excels in Florianopolis, offering a 64.5 % reduction in power expenses (equivalent to \$211). Following closely are Sao Paulo (28 % reduction in electricity expenditure, equivalent to \$60) and Nanjing (37.9 % reduction in electricity expenditure, equivalent to \$47). However, for Rende's room samples, XGB-DQN may increase HVAC usage to maintain occupant comfort, resulting in an uptick in the electricity bill. Despite a modest 9.6 % increase in the proportion of electricity expenditure, the final bill rises by \$107 due to the expensive local electricity prices.

Table 5

Comparison of comfort duration, energy consumption, and electricity cost differences among models.

Indicators	Scenario	Rende	Florianopolis	Sao paulo	Nanjing	Total
Total Records (h)	\	4128.00	768.00	12732.00	900.00	18528.00
HVAC operating time (h)	Baseline	0.00	0.00	0.00	0.00	0.00
	Scenario-1	592.00	227.00	2451.00	231.00	3501.00
	Scenario-2	649.00	81.00	1762.00	144.00	2636.00
Window opening time (h)	Baseline	0.00	0.00	0.00	0.00	0.00
	Scenario-1	508.00	293.00	8971.00	562.00	10334.00
	Scenario-2	1441.00	339.00	4476.00	329.00	6585.00
Duration of thermal comfort (h)	Baseline	2571.00	561.00	9119.00	517.00	12768.00
	Scenario-1	2599.00	613.00	9626.00	659.00	13497.00
	Scenario-2	4040.00	735.00	11827.00	620.00	17222.00
Thermal comfort (%)	Scenario-1	0.01	0.07	0.04	0.16	0.04
	Scenario-2	0.36	0.23	0.21	0.11	0.24
Absolute Energy Consumption (kwh)	Scenario-1	888.00	195.22	2107.86	166.32	3357.40
	Scenario-2	973.50	69.66	1515.32	103.68	2662.16
Relative Energy Consumption (kw)	Scenario-1	0.34	0.32	0.22	0.25	0.28
	Scenario-2	0.24	0.09	0.13	0.17	0.16
Estimated electricity cost (\$)	Scenario-1	1113.70	327.36	213.19	124.68	414.25
	Scenario-2	1220.93	116.80	153.26	77.67	385.95

Notes: Scenario-1 represents the case of occupant spontaneous control, while Scenario-2 represents the case controlled by the XGB-DQN program.

4.3. Collaborative control of HVAC and windows

XGB-DQN intelligently coordinates the control of HVAC and window states based on the dynamic changes in indoor and outdoor environments to minimize heat loss or reduce energy waste. Fig. 7 illustrates the heat-map of window opening probabilities under Occupant controlled operations and XGB-DQN controlled operations. It can be observed that when the indoor temperature falls below the thermal comfort range, the probability of XGB-DQN opening the windows is significantly lower than under occupant autonomous control, thereby reducing heat loss. Similarly, when the indoor temperature exceeds the comfort range, the XGB-DQN algorithm triggers air conditioning for cooling and simultaneously closes the windows to avoid energy waste caused by air circulation. Additionally, during periods of thermal comfort, XGB-DQN can flexibly adjust the opening and closing of windows, facilitating effective air exchange between the indoor and outdoor environments.

4.4. Weekly case analysis

Fig. 8 presents the control performance of the XGB-DQN model on indoor environment for room samples in four cities over a continuous time period. The four lines in the graph depict the actual outdoor temperature variation, actual indoor temperature variation, predicted trend of indoor temperature variation under XGB-DQN control, and predicted trend of indoor temperature variation under the baseline model. The gray background represents the expanded adaptive thermal comfort zone. The HVAC system states under actual conditions and XGB-DQN control are listed, where 0 indicates off, and 1 indicates on. At midnight each day, the XGB-DQN algorithm's initial indoor state is reset based on the real environment.

Initially, XGB-DQN demonstrates the capacity to regulate indoor temperatures within the thermal comfort zone during working hours. As shown in the graph, when the indoor environment fails to meet thermal comfort requirements during working hours, XGB-DQN initiates cooling or heating. Consequently, within one to 2 h of operation, the indoor temperature swiftly adjusts to the thermal comfort zone, providing occupants with a comfortable office environment. XGB-DQN can concurrently consider both the short-term and long-term benefits of behavior, ensuring the indoor environment consistently maintains comfort. In comparison, occupant control behavior may indeed improve the indoor environment in the short term. However, improper or prolonged operation can lead to a rapid temperature decrease, surpassing the comfort range, causing discomfort and resource wastage.

Additionally, the line chart illustrating changes in HVAC activation

status indicates that XGB-DQN has met its energy-saving objectives. In comparison to occupant-initiated control, XGB-DQN achieves the required indoor temperature with a shorter runtime for the HVAC system. This implies that XGB-DQN promptly shuts down the HVAC system upon detecting a stable indoor environment, effectively reducing unnecessary energy consumption. Moreover, following the reward mechanism of XGB-DQN, windows primarily remain closed during air conditioning operation to prevent heat loss/gain from convective effects, unless enhanced ventilation can improve cooling or heating efficiency.

5. Discussion

5.1. The utilization of ASHRAE global occupant behavior database

This study represents the inaugural utilization of occupant behavior and environmental data from multiple countries within the ASHRAE Global Building Occupant Behavior Database [9]. Previous research predominantly relied on control methods such as rule-based control set by experts [29,30], model predictive control utilizing thermodynamic models [31–33], and simulation techniques on platforms like Energy Plus [8,51] due to the lack of on-site monitoring of occupant behavior and environmental variations. However, scholars have identified limitations in these models, such as difficulties in controlling dynamic and intricate environments, the need for a sophisticated understanding of physical model foundations, and discrepancies with real-world scenarios, making their practical application in engineering challenging [41].

As such, this study aims to predict and analyze long-term data on measured building occupant behavior and indoor/outdoor environmental conditions using the ASHRAE Global Building Occupant Behavior Database. The data set encompasses occupant interactions with various building components, including HVAC systems, windows, and doors. This allows reinforcement learning models to coordinate control across various building elements, offering opportunities to achieve dual objectives of thermal comfort and energy conservation. Simultaneously, the data set provides real-world data from multiple countries and climatic zones, serving as a robust foundation and reference for predicting the influence of occupant behavior on indoor environments.

To ensure data consistency and usability, this study introduces a standardized data preprocessing paradigm: (1) filtering of features and selecting datasets containing the required variables; (2) standardizing the time step across all datasets; (3) removing missing values and

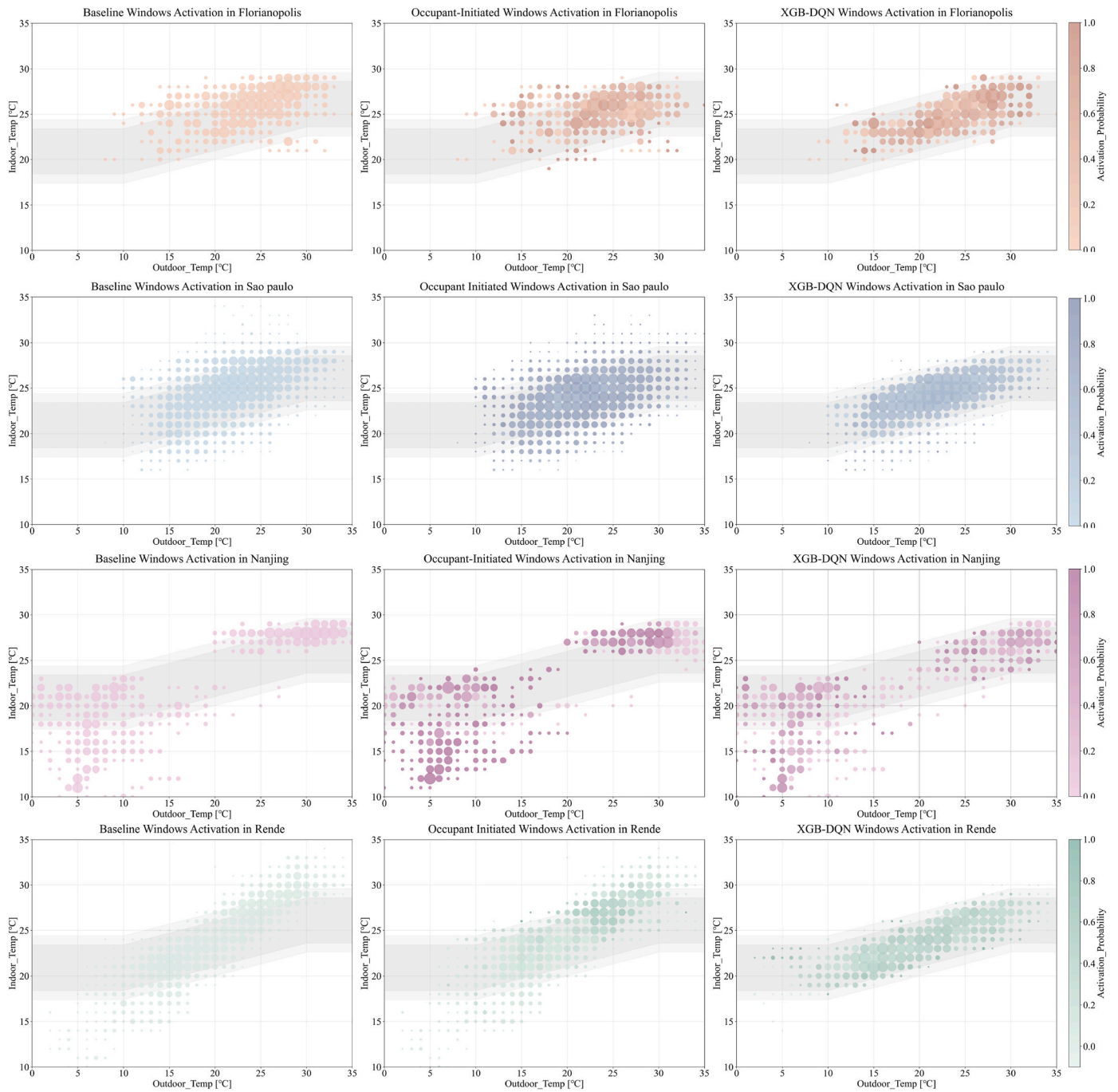


Fig. 7. Heat-maps of window opening probability for room samples in four cities under three models.

rectifying data records conflicting with the original data set; (4) constructing feature engineering to expand and complement the data. The results demonstrate the accurate prediction of temperature changes at the next time step based on indoor and outdoor environmental conditions and occupant behavior, with errors within $0.18\text{ }^{\circ}\text{C}$, indicating a well-fitted model.

Nonetheless, this study has certain limitations. Firstly, as only five datasets include data on indoor-outdoor environmental conditions, HVAC, and window states, this paper selectively focused on room samples from Nanjing, Florianopolis, Sao Paulo, and Rende. The generalizability of the research findings to other building types and cities has yet to be validated. Secondly, the study targeted HVAC and window components with the highest correlation to indoor temperature changes as control objectives, overlooking other building elements. A

more comprehensive consideration of other building elements related to indoor thermal environment could further enhance the accuracy of predicting indoor environmental changes. Lastly, given that the ASH-RAE occupant data set lacks recorded information on occupants' presence status and numbers [66], this study set the initiation time of XGB-DQN by establishing a conventional work schedule. However, identifying features indicative of occupant presence could enable more refined HVAC and window control [73], thereby enhancing the scientific precision of building equipment control decisions.

5.2. Application of XGBoost for anticipating occupant behaviors in indoor thermal conditions

This study puts forth a series of methods to enhance the accuracy of

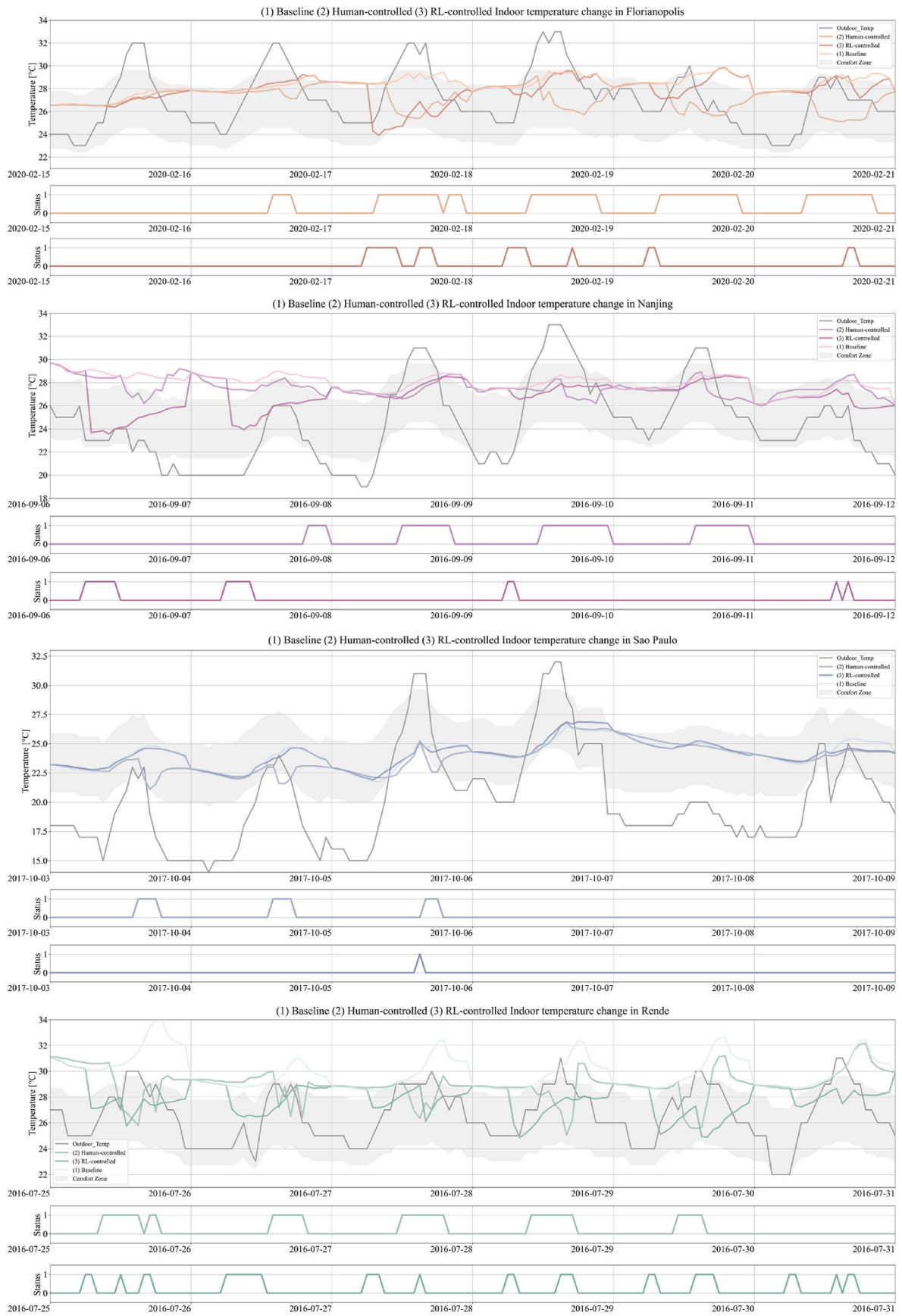


Fig. 8. Comparison of the application effects of the XGBoost-DQN model, baseline model, and resident control model on room samples in four cities over the course of one week.

state change predictions in reinforcement learning through the application of the XGBoost algorithm. Firstly, variables with strong time dependence and lag should no longer be considered as features for predicting the state at the next time step in the XGBoost method. Similarly, if the predicted variable exhibits coordinated changes with the variables used for prediction but lacks a causal relationship, it is not suitable as a prediction target. For instance, scholars have highlighted that in LSTM models, using variables like current indoor temperature and insulation layer temperature to predict the next time step's indoor air temperature may lead to convergence issues and poor model fitting [33]. To address this concern, the present study opts to predict the change in indoor temperature as an indirect means to forecast the temperature value at the next time step.

Additionally, the accuracy of model predictions can be elevated through appropriate feature engineering. This study employed two approaches: deducing parameters such as operational duration and cumulative operational duration from existing window and air conditioning status records in the ASHRAE database and supplementing outdoor meteorological indicators with additional open-source information. In addition to feature engineering, Bayesian methods are utilized to optimize hyper-parameters, leading to an enhancement in the accuracy of XGBoost predictions [64]. The research results indicate that, following feature expansion, the R^2 of XGBoost increased from 0.199 to 0.466, and the RMSE decreased from 0.46 to 0.35.

Finally, to ensure the effectiveness of generating continuous control policies in the reinforcement learning process, this study introduces an iterative approach to the framework of our reinforcement algorithm. Specifically, after inputting the initial state, the policy selected by the reinforcement learning is mapped to a numerical representation through a Python function. This representation is then merged with other features and fed into XGBoost for prediction. By combining the next time step's change in indoor temperature returned by XGBoost with the initial state's indoor temperature, the state at $St+1$ can be updated, thereby overcoming challenges in the iteration of time series data [33]. The research findings demonstrate that the iterative algorithm utilized in XGB-DQN accurately fits the data to real-world scenarios.

However, the XGBoost algorithm also exhibits certain limitations. Some scholars argue that Long Short-Term Memory (LSTM), a Recurrent Neural Network (RNN) based algorithm, may offer superior predictions for time series data [7]. In this study, the discontinuity in the provided data dates from the database prevented the use of LSTM for predicting intermittent data. In future research, investigators could design more comprehensive experimental procedures and explore the integration of LSTM and RL to enhance performance.

5.3. XGB-DQN achieves the goal of improving thermal comfort and energy efficiency

To assess the performance of the XGB-DQN model in terms of thermal comfort and energy efficiency, this study conducts a comparison with baseline models and occupant-driven control. The results reveal that, in comparison to the baseline model, occupant-driven control increases the duration of indoor thermal comfort by 4 %, while the XGB-DQN model demonstrates a significant improvement of 24 %. Simultaneously, while enhancing indoor thermal comfort levels, the XGB-DQN controlled air conditioning runtime decrease by 24.7 % compared to occupant-driven control. These findings are in line with other studies, such as the research by Chen et al. where the Q-learning algorithm reduced energy consumption by 23 % and decreased discomfort hours by 80 % [6]. Similarly, An et al. achieved a 16 % improvement in thermal comfort and a 23 % reduction in energy consumption through the use of Deep Q learning algorithms [42].

From a more specific perspective, this study selected rooms from four international cities as test samples to demonstrate the variations in thermal comfort, energy consumption, and electricity cost performance of XGB-DQN in different regions. The research findings indicate that

XGB-DQN exhibits the highest potential for improving thermal comfort in the Rende sample rooms (36 %). However, this improvement comes at the cost of an additional 9.6 % in electricity consumption and expenditure (equivalent to \$107 annually). In contrast, XGB-DQN shows significant energy-saving potential to reduce electricity bills in Florianopolis, achieving a simultaneous 23 % improvement in thermal comfort and a 64.5 % reduction in electricity costs (equivalent to \$211 annually). Furthermore, Florianopolis and Sao Paulo, both located on the southeastern coast of Brazil, demonstrate similar trends in enhancing thermal comfort performance (23 % and 21 %) of XGB-DQN in their respective samples. Therefore, in cities with similar geographical locations, climate characteristics, and architectural features, the performance of XGB-DQN may be more comparable. However, further validation is still required through additional samples and experimental data.

However, the construction of the XGB-DQN algorithm also presents certain limitations. The reward function integrates considerations for both indoor thermal comfort and energy consumption but includes the air conditioning runtime multiplied by a fixed energy consumption per unit time as the measure of energy consumption. In reality, the actual energy consumption of air conditioning is influenced by factors such as different operating phases, indoor-outdoor temperature differentials, and the difference between indoor temperature and the set temperature [74]. These factors are not accounted for in the current formulation. Future research could benefit from the utilization of more precise energy measurement tools [75] and behavior monitoring systems to enhance the dimensionality and quality of the data. Additionally, experiments and applications in real building spaces could be explored to more accurately assess the performance of RL models in practical scenarios.

6. Conclusion

This study introduces a reinforcement learning algorithm that combines Deep Q Network (DQN) and XGBoost, aiming to achieve the collaborative control of HVAC and windows. The model demonstrates the ability to coordinate the control of HVAC and windows, resulting in the overall optimization of building systems, which can help to measure the improvement of thermal comfort and energy saving. Several conclusions are derived from this study.

- XGBoost demonstrates precision in predicting state changes in subsequent time steps during the reinforcement learning process, with a model fit accuracy represented by an R^2 of 0.466 and an RMSE of 0.353. When applied in generating DQN strategies over a continuous time period, the predictive results of XGBoost align closely with real data;
- In comparison to baseline models and occupant spontaneous control models, XGB-DQN not only significantly increases the duration of indoor thermal comfort (by 24 %) but also effectively reduces air conditioning runtime (by 24.7 %), achieving the dual goals of thermal comfort and energy conservation;
- The research discusses the performance differences of the XGB-DQN model in room samples across various international cities. Specifically, sample rooms in Rende exhibit the greatest potential for improvement in thermal comfort performance, while those in Florianopolis have the highest energy-saving potential.

Future research could explore additional building components related to indoor thermal comfort, enhancing reinforcement learning models through their collaborative control. Moreover, there is potential to develop more refined and personalized occupant thermal comfort control strategies within different types of building spaces to increase the model's applicability in real-world scenarios.

Data and code availability statement

The data is available at: <https://ashraeobdatabase.com/> and the relevant code is available at <https://github.com/WHU-LX/Hvac-Window-based-XGB-DQN>.

CRedit authorship contribution statement

Xin Liu: Writing – original draft, Visualization, Methodology, Investigation, Data curation. **Zhonghua Gou:** Writing – review & editing, Supervision, Project administration, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix A. Summary Table of Literature Review

Table A1

Reinforcement Learning for Enhanced Indoor Thermal Comfort: A Literature Review

ID	Site	Type	Start-End	Interval	Independent Variables	Reward	Target	Methodology	Results
[7]	Beijing, China	Office	2015 03.16.-05.15	10min	Outdoor Temperature, Solar Radiation, Air Quality, Wind Speed, Wind Direction, Human Wavelength		Window Status	RNN-LSTM, Q-learning, SARSA	Improves thermal and indoor air quality by more than 90 %
[8]	America	Commercial	01.01–04.01	15min	Indoor Temperature, Humidity, Solar Radiation, Wind Speed, Wind Direction, Hours, Occupancy	PPD, Energy	HVAC set point	Dirichlet change point algorithm, Deep Q Network, Energy Plus	Achieve energy savings of up to 13 % and improve thermal comfort by up to 9 %.
[43]	\	ASHRAE RP884	\	\	Mean Air Temperature, Relative Humidity, Metabolic Rate, Clothing Insulation, Radiant Temperature, Maximum Temperature	PMV	HVAC set point	Gaussian Mixture Model, K-Nearest Neighbors, Support Vector Machine, Random Forest, Q-Learning	The recall rate of the thermal comfort model is 49.3 %.
[76]	America	Residential	2019–2020	60min	Outdoor Temperature, Indoor Temperature, Comfort Threshold, Retail Electricity Price	Cost, Temperature Violations	HVAC Status	Deep Deterministic Policy Gradient	Reduce energy costs by 15 % and decrease thermal comfort violations by 79 %.
[6]	Miami and Los Angeles, America	Residential	One year	60min	Outdoor Temperature, Solar Radiation, Wind Speed, Indoor Temperature, Total Heat Gain	Electricity Consumption, Thermal Comfort	HVAC-Window Status	Heuristic Control, Q-learning	Reduces energy consumption by 13 % and 23 %, decrease discomfort hours by 62 % and 80 %.
[77]	Purdue, America	Residential	2018 9 months	15 min	Indoor Temperature, Indoor Humidity, Thermostat Set-point, Clothing Level, Metabolic Rate, Room Occupancy Rate, Time of Day	TSV	Thermostat Set point, Clothing Behavior	Q-learning, Energy plus	The combination of RL and transfer learning enables accurate prediction of occupants' behavior in buildings with an R2 greater than 0.73.
[42]	Hongkong, China	Residential	2022 07.17–08.08	1min	Indoor PM2.5 Concentration, Outdoor PM2.5 Concentration, Indoor Temperature, Outdoor Temperature, Hours	Indoor PM2.5, Thermal Comfort, Energy Consumption	Windows, Air Conditioning, Air Purifier	Deep Q Learning	Health and thermal comfort have improved by 21 % and 16 %, respectively, with a 23 % reduction in energy consumption.

(continued on next page)

Table A1 (continued)

ID	Site	Type	Start-End	Interval	Independent Variables	Reward	Target	Methodology	Results
[4]	Changsha, China	Apartment	May–September	30min	Indoor Temperature, Outdoor Temperature, Electricity Price, Number of Occupants, Time Step Length	Energy Consumption, Indoor Thermal Comfort, Room Occupancy Rate, Electricity Price	HVAC Status	Markov Decision Process, Deep Q Network	ED-DQN achieves outstanding performance by minimizing energy consumption while maintaining thermal comfort to the maximum extent.
[40]	Singapore	Office	10,000 Hours	60min	Indoor Temperature, Outdoor Temperature, Indoor Humidity, Outdoor Humidity	Thermal Comfort, Energy Consumption	Temperature and Humidity set point	DDPG, SARSA, DQN, Q-learning	Improve occupants' thermal comfort by 13.6 %, and result in 4.31 % reduction in energy consumption.
[33]	Beijing, China	Residential	2020–2021 11.06–03.20	60min	Indoor Temperature, Fresh Air Temperature, Exhaust Air Temperature, Return Air Temperature, Electricity Consumption	Thermal Comfort, Energy Consumption	HVAC set point	XGBoost, LSTM, DQN, DDPG	DDPG ensures indoor thermal comfort and achieves a 10.06 % energy savings.

Appendix B. Temporal Datasets and Climatic Impact on Adaptive Thermal Comfort Zone

Table B1

Dataset Information

ID	City	Climate	Room_Function	Room_ID	Records	Publications
6	Florianopolis, Brazil	Cfa	Educational Office	A	1693	Unpublished, by Mateus Bavaresco
10	Rende, Italy	Cfa	Educational Office	B	9461	Mora et al. [48]
16	Sao Paulo, Brazil	Cfa	Commercial Office	H–R	22,339	Unpublished, by Karin Chvatal
21	Nanjing, China	Cfa	Commercial Office	C, D	1818	Unpublished, by Xin Zhou, Jiawen Ren
23	Sao Paulo, Brazil	Cfa	Commercial Office	E–G	8640	Neves et al. [47]

The data for Florianopolis spans from February 2019 to December 2020, while Rende's data covers the period from May 2016 to May 2017. The data for Nanjing is sporadic, distributed between June and September 2017 and from December 2017 to June 2018. Room records for Sao Paulo are more extensive, with data ranging widely from January 2018 to December 2019. The room IDs and the temporal heat-map of the data set are presented below in Fig B1.

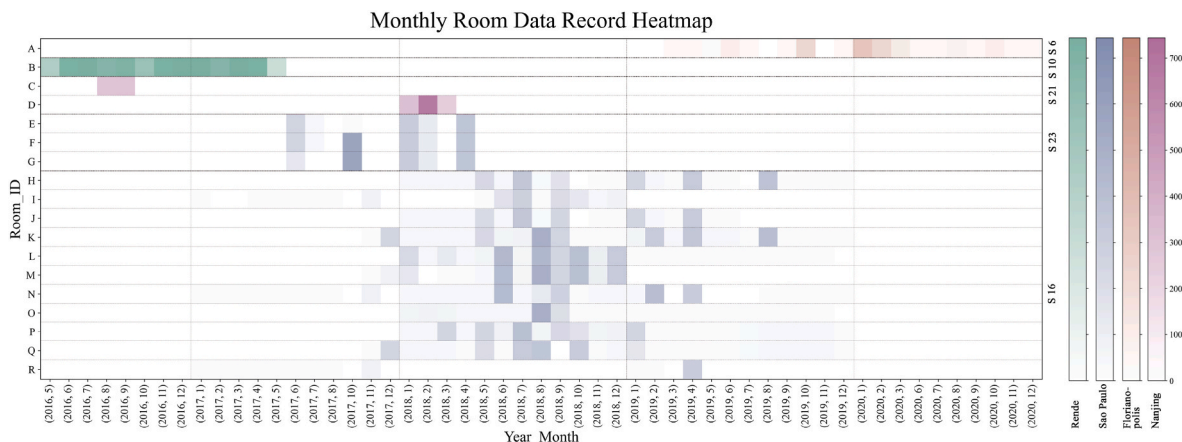


Fig. B1. Heatmap of Temporal Distribution for Five Datasets.

The climate characteristics and corresponding adaptive thermal comfort ranges for the four cities are as follows: (1) Nanjing (Cfa), situated in the southeast coastal region of China, experiences a subtropical monsoon climate with distinct seasons and significant temperature variations. Summer temperatures in July and August can reach as high as 38 °C, while winter temperatures may drop to minus 10 °C. Consequently, approximately half of the year sees temperatures in Nanjing lying outside the adaptive thermal comfort range. (2) Rende (Csa), located in southern Italy near Cosenza, has a Mediterranean climate characterized by hot, dry summers and mild, rainy winters. During most of the summer, temperatures are within the thermal comfort range, while in winter, temperatures generally fall below it. (3) Florianopolis (Cfa), situated on the southern coast of Brazil, shares a climate

similar to Sao Paulo. Summers are hot and rainy, with temperatures mostly within the thermal comfort range, while winters experience slightly lower temperatures outside the thermal comfort zone. (4) Sao Paulo (Cfa), in the southeastern part of Brazil and closer to the Tropic of Capricorn, has a subtropical humid climate. With a higher average annual temperature and minimal temperature fluctuations, Sao Paulo maintains temperatures around 20 °C for the majority of the year, within the thermal comfort range. The outdoor temperature trends for the four cities and their corresponding adaptive thermal comfort zone ranges are illustrated in Figure B2.

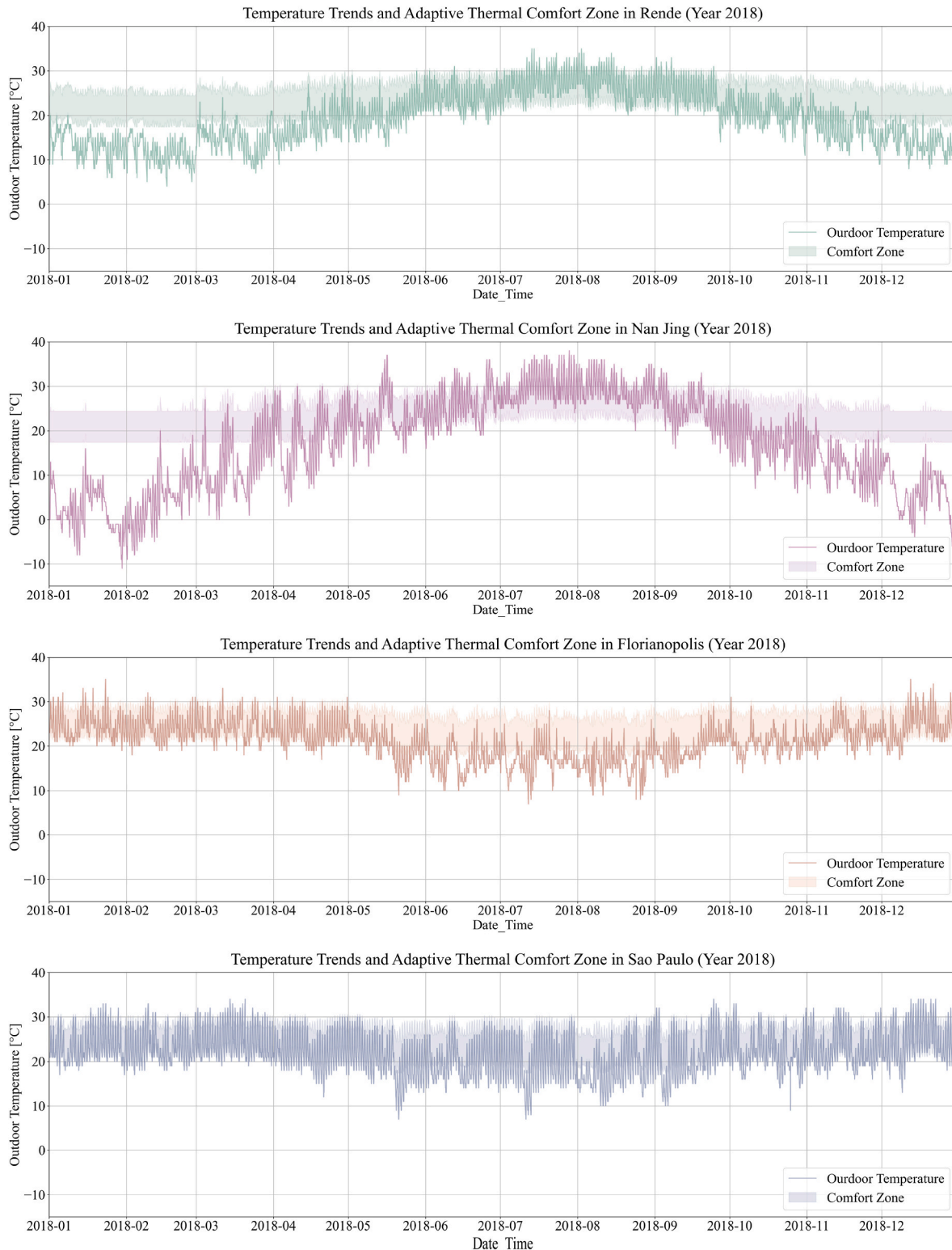


Fig. B2. Trends in Outdoor Temperature Variation for Four Cities and Corresponding Adaptive Thermal Comfort Zone.

Appendix C. Summary Table of Reward Function

Table C2

Summary of Reward Function Settings in Reinforcement Learning Models

ID	Target	Variables	Definition
[66]	$r(t) = -\beta(EC(T)) - (1 - \beta)O(t)(T_{op}(t) - T_{sp})^2$ \ Occupancy Electricity consumption Thermal comfort	β $O(t)$ EC T_{op} T_{sp}	Coefficient to weight energy and thermal comfort Binary occupancy state Amount of energy consumption in kWh Indoor operative temperature Desired set point temperature
[6]	$L = \alpha(p \sum B_{AC}) + \beta(\sum (T_{in} - T_{op})^2) + \gamma(\sum (RH_{in} - RH_{op})^2)$ \ Electricity consumption Thermal comfort Humidity comfort	α, β, γ p B_{AC} T_{in} T_{op} RH_{in} RH_{op}	weights that are subject to the user's priority Constant rate Boolean value that represents HVAC operation Indoor temperature Target temperature Indoor relative humidity Target relative humidity
[42]	$R(t) = -\left(\omega_1 \frac{C_{in}^t}{C_{norm}} + \omega_2 \frac{ T_{in}^t - T_{set} }{T_{up} - T_{lo}} + \omega_3 \frac{P_{cl} + P_{co}}{P_{cl,r} + P_{co,r}}\right)$ \ Indoor PM2.5 Thermal comfort Electricity consumption	$\omega_1, \omega_2, \omega_3$ C_{in}^t C_{norm} T_{in}^t and T_{set} T_{up} and T_{lo} P_{cl} and P_{co} $P_{cl,r}$ and $P_{co,r}$	Weights for PM2.5 comfort and energy, sum = 1 Indoor PM2.5 concentration Reference PM2.5 concentration for normalization Indoor temperature and set point temperature Upper and lower bound temperatures Power consumption of air cleaner & air conditioner Rated power of the air cleaner & the air conditioner
[77]	$R = TSV_t - TSV_{t+1} , \text{ or } R = PMV_t - PMV_{t+1} $ Thermal comfort	TSV PMV	Thermal sensation vote Predicted mean vote
[78]	$r_t = -C_{in}(t + \Delta t)$ Indoor PM2.5	C_{in}	Indoor PM2.5 concentration
[38]	$R(t) = -\left(\frac{VSL \bullet Mort(C_{in}(t + \Delta t)) \bullet p}{D \bullet M \bullet p} + Price \bullet Power \bullet AC(t) \bullet T\right)$ Economic loss due to PM2.5-related diseases Economic loss due to energy consumption	VSL $Mort()$ $C_{in}(t + \Delta t)$ p, D, M $Price$ $Power$ T	Statistical life, set at 1,980,000 CNY per death Yearly mortality attributable to C_{in} environment Indoor PM2.5 concentration Population, 365 days, 1440min per day Price of electricity, set at 0.5 CNY/kWh for China Power when air cleaner kept on, set at 10.1CADR/V 1/60 h per minute
[79]	$R(t) = -\left(\partial_1 \times \frac{PM(T) \times PM(T)}{PM(0) \times PM(0)} + \partial_2 \times R_{energy}\right)$ \ Air Quality Energy consumption	∂_1, ∂_2 $PM0$ R_{energy}	Weighted coefficients Particulate matter with a diameter Energy consumption
[33]	$R = -\lambda \times \eta_1 \times (f(E_t - E_{min})) + \eta_2 \times \sum_{i=1}^n (T_i - T_{exp})^2$ \ Energy consumption Thermal comfort	λ, η_1, η_2 f E_{min} T_i T_{exp}	Amplification coefficient, $\lambda = 1$, conversion factors Activation function to facilitate the RL convergence Predicted minimum energy consumption value Indoor temperature of three bedrooms Desired indoor temperature
[4]	$r(t) = \sum_i \left[(E_{off}(t) - \alpha E_{HVAC}(t) - SW_{penalty}(t)) + \beta \left(\frac{1}{1 + e^{(T_{in} - th_{best})^2}} - R_{penalty}(t) \right) \right]$ \ Thermal comfort Energy consumption	α, β T_{in} th_{best} $R_{penalty}(t)$ $E_{off}(t)$ $E_{HVAC}(t)$ $SW_{penalty}(t)$	Weights to balance energy and thermal comfort Indoor temperature The most comfortable indoor temperature 0 when T_{in} between T_{max} and T_{min} , or $ T_{in} - T_m $ Reward when shutting down HVAC, else 0 Energy consumption of HVAC systems in each zone The cost of switching the HVAC system
[8]	$r(t) = -\omega_e P_{HVAC}(t) - \omega_c C_{comfort}(t) OCC$ \ Thermal comfort Electricity consumption	ω_e, ω_c $C_{comfort}(t)$ OCC $P_{HVAC}(t)$	Weights for the electric demand (0.4) and thermal comfort (0.6) Equals PPD when PPD falls below threshold, else 1 Occupant status flag, 0 when no occupancy, else 1 Total HVAC energy consumption

References

- [1] UNEP, 2022 Global Status Report for Buildings and Construction, 2022. <https://www.unep.org/resources/publication/2022-global-status-report-buildings-and-construction>. (Accessed 24 November 2023).
- [2] IEA, World Energy Outlook 2022, 2022. <https://www.iea.org/reports/world-energy-outlook-2022>. (Accessed 24 November 2023).
- [3] M. Esrafilian-Najafabadi, F. Haghighat, Towards self-learning control of HVAC systems with the consideration of dynamic occupancy patterns: application of model-free deep reinforcement learning, *Build. Environ.* 226 (2022) 15.
- [4] Q.M. Fu, Z. Li, Z.K. Ding, J.P. Chen, J. Luo, Y.Z. Wang, Y. Lu, ED-QDN: an event-driven deep reinforcement learning control method for multi-zone residential buildings, *Build. Environ.* 242 (2023) 17.
- [5] D. Weinberg, Q. Wang, T.O. Timoudas, C. Fischione, A review of reinforcement learning for controlling building energy systems from a computer science perspective, *Sustain. Cities Soc.* 89 (2023) 12.
- [6] Y.J. Chen, L.K. Norford, H.W. Samuelson, A. Malkawi, Optimal control of HVAC and window systems for natural ventilation through reinforcement learning, *Energy Build.* 169 (2018) 195–205.
- [7] M.J. Han, R. May, X.X. Zhang, X.R. Wang, S. Pan, D. Yan, Y. Jin, A novel reinforcement learning method for improving occupant comfort via window opening and closing, *Sustain. Cities Soc.* 61 (2020) 11.
- [8] X.T. Deng, Y. Zhang, H. Qi, Towards optimal HVAC control in non-stationary building environments combining active change detection and deep reinforcement learning, *Build. Environ.* 211 (2022) 16.
- [9] B. Dong, Y.P. Liu, W. Mu, Z.X. Jiang, P. Pandey, T.Z. Hong, B. Olesen, T. Lawrence, Z. O'Neill, C. Andrews, E. Azar, K. Bandurski, R. Bardhan, M. Bavaresco, C. Berger, J. Burry, S. Carlucci, K. Chvatal, M. De Simone, S. Erba, N. Gao, L.T. Graham, C. Grassi, R. Jain, S. Kumar, M. Kjergaard, S. Korsavi, J. Langevin, Z.R. Li, A. Lipczynska, A. Mahdavi, F. Malik, M. Marschall, Z. Nagy, L. Neves, W. O'Brien, S. Pan, J.Y. Park, I. Pigliautile, C. Piselli, A.L. Pisello, H.N. Rafsanjani, R.F. Rupp, F. Salim, S. Schiavon, J. Schwee, A. Sonta, M. Touchie, A. Wagner, S. Walsh, Z. Wang, D.M. Webber, D. Yan, P. Zangheri, J.S. Zhang, X. Zhou, X. Zhou, A global building occupant behavior database, *Sci. Data* 9 (1) (2022) 15.
- [10] A. Ashrae, ASHRAE Standard 55-2017: Thermal Environmental Conditions for Human Occupancy, American Society of Heating, Refrigerating and Air-Conditioning Engineers, Atlanta, GA, USA, 2017.
- [11] P.O. Fanger, Thermal Comfort, Analysis and Application in Environmental Engineering, McGraw Hill, New York, 1972.
- [12] O.V. Semko, O.I. Filonenko, O.I. Yurin, N.M. Mahas, A.R. Norka, Analytical Determination and Interpretation of Thermal Comfort, Publishing House "Baltija Publishing", 2022.
- [13] S. Carlucci, L. Bai, R. de Dear, L. Yang, Review of adaptive thermal comfort models in built environmental regulatory documents, *Build. Environ.* 137 (2018) 73–89.
- [14] J.F. Nicol, M.A. Humphreys, Thermal Comfort as Part of a Self-Regulating System, 1973.
- [15] M.A. Humphreys, Field Studies of Thermal Comfort Compared and Applied, 1976.
- [16] J.F. Nicol, M.A. Humphreys, Adaptive thermal comfort and sustainable thermal standards for buildings, *Energy Build.* 34 (6) (2002) 563–572.
- [17] R.J. De Dear, G.S. Brager, J. Reardon, F. Nicol, Developing an adaptive model of thermal comfort and preference/discussion, *Build. Eng.* 104 (1998) 145.
- [18] A. Ashrae, ASHRAE Standard 55-2013: Thermal Environmental Conditions for Human Occupancy, American Society of Heating, Refrigerating and Air-Conditioning Engineers, Atlanta, GA, USA, 2013.
- [19] T. Ahmed, P. Kumar, L. Mottet, Natural ventilation in warm climates: the challenges of thermal comfort, heatwave resilience and indoor air quality, *Renewable Sustainable Energy Rev.* 138 (2021) 17.
- [20] Y.M. Liu, S.C. Liu, S.X. Wang, B. Zhao, How will window opening change under global warming: a study for China residence, *Build. Environ.* 209 (2022) 10.
- [21] A. Ashrae, ASHRAE Standard 55-2020: Thermal Environmental Conditions for Human Occupancy, American Society of Heating, Refrigerating and Air-Conditioning Engineers, Atlanta, GA, USA, 2020.
- [22] M.H. Luo, B. Cao, J. Damiens, B.R. Lin, Y.X. Zhu, Evaluating thermal comfort in mixed-mode buildings: a field study in a subtropical climate, *Build. Environ.* 88 (2015) 46–54.
- [23] H.Y. Yan, Z. Sun, F.N. Shi, G.D. Yuan, M.R. Dong, M.L. Wang, Thermal response and thermal comfort evaluation of the split air conditioned residential buildings, *Build. Environ.* 221 (2022) 17.
- [24] G.Y. Yun, J.H. Lee, K. Steemers, Extending the applicability of the adaptive comfort model to the control of air-conditioning systems, *Build. Environ.* 105 (2016) 13–23.
- [25] D. Sánchez-García, J. Martínez-Crespo, U.R.R. Hernando, C. Alonso, A detailed view of the Adaptive-Comfort-Control-Implementation Script (ACCIS): the capabilities of the automation system for adaptive setpoint temperatures in building energy models, *Energy Build.* 288 (2023) 15.
- [26] D. Enescu, A review of thermal comfort models and indicators for indoor environments, *Renewable Sustainable Energy Rev.* 79 (2017) 1353–1379.
- [27] T.Z. Hong, S.C. Taylor-Lange, S. D'Oca, D. Yan, S.P. Corgnati, Advances in research and applications of energy-related occupant behavior in buildings, *Energy Build.* 116 (2016) 694–702.
- [28] P.W. Tien, S.Y. Wei, J.K. Calauti, J. Darkwa, C. Wood, Real-time monitoring of occupancy activities and window opening within buildings using an integrated deep learning-based approach for reducing energy demand, *Appl. Energy* 308 (2022) 25.
- [29] M. Fiorentini, G. Serale, G. Kokogiannakis, A. Capozzoli, P. Cooper, Development and evaluation of a comfort-oriented control strategy for thermal management of mixed-mode ventilated buildings, *Energy Build.* 202 (2019) 16.
- [30] S. Firlag, M. Yazdani, C. Curcija, C. Kohler, S. Vidanovic, R. Hart, S. Czarnecki, Control algorithms for dynamic windows for residential buildings, *Energy Build.* 109 (2015) 157–173.
- [31] B. Pandey, B. Bohara, R. Pungaliya, S.C. Patwardhan, R. Banerjee, A thermal comfort-driven model predictive controller for residential split air conditioner, *J. Build. Eng.* 42 (2021) 18.
- [32] D. Kim, J.E. Braun, Development, implementation and performance of a model predictive controller for packaged air conditioners in small and medium-sized commercial building applications, *Energy Build.* 178 (2018) 49–60.
- [33] M. Wang, B.R. Lin, MF²: model-free reinforcement learning for modeling-free building HVAC control with data-driven environment construction in a residential building, *Build. Environ.* 244 (2023) 21.
- [34] R.S. Sutton, A.G. Barto, Introduction to Reinforcement Learning, MIT press Cambridge, 1998.
- [35] M. Minsky, Steps toward artificial intelligence, *Proc. IRE* 49 (1) (1961) 8–30.
- [36] V. Mnih, K. Kavukcuoglu, D. Silver, A. Graves, I. Antonoglou, D. Wierstra, M. Riedmiller, Playing Atari with Deep Reinforcement Learning, 2013 arXiv preprint arXiv:1312.5602.
- [37] V. Mnih, K. Kavukcuoglu, D. Silver, A.A. Rusu, J. Veness, M.G. Bellemare, A. Graves, M. Riedmiller, A.K. Fidjeland, G. Ostrovski, Human-level control through deep reinforcement learning, *Nature* 518 (7540) (2015) 529–533.
- [38] Y.T. An, Z.L. Niu, C. Chen, Smart control of window and air cleaner for mitigating indoor PM_{2.5} with reduced energy consumption based on deep reinforcement learning, *Build. Environ.* 224 (2022) 10.
- [39] K. Dalamagkidis, D. Kolokotsa, K. Kalaitzakis, G.S. Stavrakakis, Reinforcement learning for energy conservation and comfort in buildings, *Build. Environ.* 42 (7) (2007) 2686–2698.
- [40] G.Y. Gao, J. Li, Y.G. Wen, DeepComfort: energy-efficient thermal comfort control in buildings via reinforcement learning, *IEEE Internet Things J.* 7 (9) (2020) 8472–8484.
- [41] H. Mo, H.J. Sun, J.J. Liu, S. Wei, Developing window behavior models for residential buildings using XGBoost algorithm, *Energy Build.* 205 (2019) 8.
- [42] Y.T. An, C. Chen, Energy-efficient control of indoor PM_{2.5} and thermal comfort in a real room using deep reinforcement learning, *Energy Build.* 295 (2023) 11.
- [43] S.L. Lu, W.L. Wang, C.C. Lin, E.C. Hameen, Data-driven simulation of a thermal comfort-based temperature set-point control with ASHRAE RP884, *Build. Environ.* 156 (2019) 137–146.
- [44] Y.P. Liu, B. Dong, T.Z. Hong, B. Olesen, T. Lawrence, Z. O'Neill, Ashrae URP-1883: development and analysis of the ASHRAE global occupant behavior database, *Sci. Technol. Built Environ.* 29 (8) (2023) 749–781.
- [45] N. Luo, G. Fierro, Y.P. Liu, B. Dong, T.Z. Hong, Extending the Brick schema to represent metadata of occupants, *Autom. Constr.* 139 (2022) 13.
- [46] R.F. Rupp, R.K. Andersen, J. Toftum, E. Ghisi, Occupant behaviour in mixed-mode office buildings in a subtropical climate: beyond typical models of adaptive actions, *Build. Environ.* 190 (2021) 107541.
- [47] L.O. Neves, A.P. Hopes, W.J. Chung, S. Natarajan, "Mind reading" building operation behaviour, *Energy Sustain. Dev.* 56 (2020) 1–18.
- [48] D. Mora, G. Fajilla, M.C. Austin, M. De Simone, Occupancy patterns obtained by heuristic approaches: cluster analysis and logical flowcharts. A case study in a university office, *Energy Build.* 186 (2019) 147–168.
- [49] J.H. Schwee, A. Johansen, B.N. Jorgensen, M.B. Kjaergaard, C.G. Mattera, F. C. Sangogboye, C. Veje, Room-level occupant counts and environmental quality from heterogeneous sensing modalities in a smart building, *Sci. Data* 6 (2019) 11.
- [50] N. Gao, M. Marschall, J. Burry, S. Watkins, F.D. Salim, Understanding occupants' behaviour, engagement, emotion, and comfort indoors with heterogeneous sensors and wearables, *Sci. Data* 9 (1) (2022) 16.
- [51] M.F. Touchie, K.D. Pressnail, Using suite energy-use and interior condition data to improve energy modeling of a 1960s MURB, *Energy Build.* 80 (2014) 184–194.
- [52] C. Piselli, A.L. Pisello, Occupant behavior long-term continuous monitoring integrated to prediction models: impact on office building energy performance, *Energy* 176 (2019) 667–681.
- [53] A. Mahdavi, C. Berger, F. Tahmasebi, M. Schuss, Monitored data on occupants' presence and actions in an office building, *Sci. Data* 6 (1) (2019) 290.
- [54] M. Schweiker, M. Kleber, A. Wagner, Long-term monitoring data from a naturally ventilated office building, *Sci. Data* 6 (1) (2019) 293.
- [55] A. Das, M.K. Annaqeeb, E. Azar, V. Novakovic, M.B. Kjaergaard, Occupant-centric miscellaneous electric loads prediction in buildings using state-of-the-art deep learning methods, *Appl. Energy* 269 (2020) 115135.
- [56] J.Y. Park, T. Dougherty, H. Fritz, Z. Nagy, LightLearn: an adaptive and occupant centered controller for lighting based on reinforcement learning, *Build. Environ.* 147 (2019) 397–414.
- [57] Z. Wang, T. Hong, M.A. Piette, M. Pitroni, Inferring occupant counts from Wi-Fi data in buildings through machine learning, *Build. Environ.* 158 (2019) 281–294.
- [58] H.N. Rafsanjani, C.R. Ahn, J.Y. Chen, Linking building energy consumption with occupants' energy-consuming behaviors in commercial buildings: non-intrusive occupant load monitoring (NIOLM), *Energy Build.* 172 (2018) 317–327.
- [59] J. Bursill, An Approach to Data-Driven Sensing and Predictive Supervisory Control for Commercial Buildings with In-Situ Evaluation, 2020.
- [60] B. Dong, Z.X. Li, G. McFadden, An investigation on energy-related occupancy behavior for low-income residential buildings, *Sci. Technol. Built Environ.* 21 (6) (2015) 892–901.
- [61] A. Sonta, T.R. Dougherty, R.K. Jain, Data-driven optimization of building layouts for energy efficiency, *Energy Build.* 238 (2021) 110815.

- [62] K. Bandurski, M. Hamerla, J. Szulc, H. Koczyk, The Influence of Multifamily Apartment Building Occupants on Energy and Water Consumption–The Preliminary Results of Monitoring and Survey Campaign, *EDP Sciences*, p. 00010..
- [63] T.Q. Chen, C. Guestrin, M. Assoc Comp, XGBoost: A Scalable Tree Boosting System, 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD), Assoc Computing Machinery, San Francisco, CA, 2016, pp. 785–794.
- [64] H.F. Lan, H.C. Hou, Z.H. Gou, A machine learning led investigation to understand individual difference and the human-environment interactive effect on classroom thermal comfort, *Build. Environ.* 236 (2023) 21.
- [65] K.Y. Sun, T.Z. Hong, A framework for quantifying the impact of occupant behavior on energy savings of energy conservation measures, *Energy Build.* 146 (2017) 383–396.
- [66] M. Esrafilian-Najafabadi, F. Haghighat, Transfer learning for occupancy-based HVAC control: a data-driven approach using unsupervised learning of occupancy profiles and deep reinforcement learning, *Energy Build.* 300 (2023) 16.
- [67] Z.Q. Wu, Y.F. Mu, S. Deng, J.J. Wang, Y.D. Bai, J. Xue, Y. Li, Y.T. Jiang, X. D. Zhang, W.C. Xu, Towards comfortable and cost-effective indoor temperature management in smart homes: a deep reinforcement learning method combined with future information, *Energy Build.* 275 (2022) 16.
- [68] Z.X. Yang, C.Q. Du, H.S. Xiao, B.Z. Li, W.X. Shi, B.L. Wang, A novel integrated index for simultaneous evaluation of the thermal comfort and energy efficiency of air-conditioning systems, *J. Build. Eng.* 57 (2022) 20.
- [69] H.S. Xiao, Z.X. Yang, W.X. Shi, B.L. Wang, B.Z. Li, Q. Song, J.B. Li, Z.K. Xu, Comparative analysis of the energy efficiency of air-conditioner and variable refrigerant flow systems in residential buildings in the Yangtze River region, *J. Build. Eng.* 55 (2022) 16.
- [70] R.E. González-Mahecha, A.F.P. Lucena, R. Garaffa, R.F.C. Miranda, M. Chávez-Rodríguez, T. Cruz, P. Bezerra, R. Rathmann, Greenhouse gas mitigation potential and abatement costs in the Brazilian residential sector, *Energy Build.* 184 (2019) 19–33.
- [71] A. Burgio, D. Cimmino, M. Dolatabadi, M. Jasinski, Z. Leonowicz, P. Siano, Virtual energy storage system for peak shaving and power balancing the generation of a MW photovoltaic plant, *J. Energy Storage* 71 (2023) 11.
- [72] Q.L. Li, Y. Zhao, F.H. Yu, A novel multichannel long short-term memory method with time series for soil temperature modeling, *IEEE Access* 8 (2020) 182026–182043.
- [73] T. Yang, A. Bandyopadhyay, Z. O'Neill, J. Wen, B. Dong, From occupants to occupants: a review of the occupant information understanding for building HVAC occupant-centric control, *Build. Simulat.* 15 (6) (2022) 913–932.
- [74] Y.C.X. Ma, C. Yu, Impact of meteorological factors on high-rise office building energy consumption in Hong Kong: from a spatiotemporal perspective, *Energy Build.* 228 (2020) 13.
- [75] S. Somwong, K. Puangsuwan, Power consumption control and monitor using IoT platform for smart office, *Przegląd Elektrotechniczny* 99 (5) (2023) 25–28.
- [76] Y. Du, H. Zandi, O. Kotevska, K. Kurte, J. Munk, K. Amasyali, E. McKee, F.X. Li, Intelligent multi-zone residential HVAC control strategy based on deep reinforcement learning, *Appl. Energy* 281 (2021) 14.
- [77] Z.P. Deng, Q.Y. Chen, Reinforcement learning of occupant behavior model for cross-building transfer learning to various HVAC control systems, *Energy Build.* 238 (2021) 15.
- [78] Y.T. An, T.L. Xia, R.Y. You, D.Y. Lai, J.J. Liu, C. Chen, A reinforcement learning approach for control of window behavior to reduce indoor PM2.5 concentrations in naturally ventilated buildings, *Build. Environ.* 200 (2021) 11.
- [79] W.Z. Shang, J.J. Liu, C.C. Wang, J.Y. Li, X.L. Dai, Developing smart air purifier control strategies for better IAQ and energy efficiency using reinforcement learning, *Build. Environ.* 242 (2023) 17.